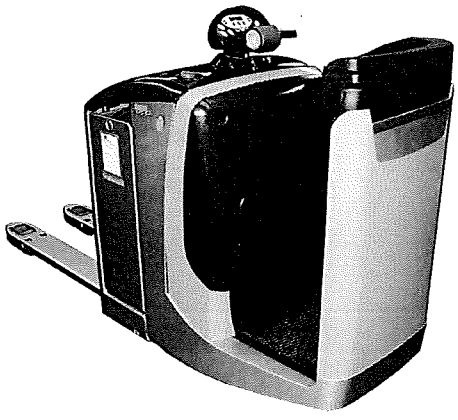
STILL

Original instructions

Electric pallet truck

EXU-S 22
EXU-S 24



CE

0177 0178

50018043464 EN - 10/2017

first in intralogistics

Address of manufacturer and contact details ▷

STILL GmbH
Berzeliusstraße 10
22113 Hamburg, Germany
Tel. +49 (0) 40 7339-0
Fax: +49 (0) 40 7339-1622
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Rules for the operating company of industrial trucks

In addition to these operating instructions, a code of practice containing additional information for the operating companies of industrial trucks is also available.

This guide provides information for handling industrial trucks:

- Information on how to select suitable industrial trucks for a particular area of application
- Prerequisites for the safe operation of industrial trucks
- Information on the use of industrial trucks
- Information on transport, initial commissioning and storage of industrial trucks

Internet address and QR code ▷

The information can be accessed at any time by pasting the address **https://m.still.de/vdma** in a web browser or by scanning the QR code.

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1

Introduction

Forklift data

We recommend that you record the principal forklift data in the following table so that they are available if required by the sales network or authorised service centre.

Type	
Serial number	
Date of delivery	

General information

- This manual contains "Original Instructions" provided by the manufacturer.
- The "operator" is defined as the person driving the forklift.
- The "user" is the physical or legal person who has the forklift truck used by the operators.
- For correct use of the forklift and in order to avoid accidents, the operator is obliged to read, understand and apply the contents of this manual, the "Rules for the use of industrial vehicles" and the labels and plates applied to the forklift.
- This manual and the attached "Rules for the use of industrial vehicles" must be kept carefully and must always be on the forklift for fast consultation.
- The manufacturer assumes no responsibility for any accidents to persons or damage to things due to the failure to observe the instructions in this manual, in the "Rules for the use of industrial vehicles" and on the labels and adhesive supplied to the forklift.
- The forklift may not be put to any use other that than indicated in this manual.
- The forklift must be used by appropriately trained operators only. For the necessary operator training, contact the authorised sales network.
- Persons working near the forklift must also be instructed in the risks associated with use of the forklift.
- In the interests of clear information, some illustrations in this manual show the forklift without the safety equipment (guards, panels, etc.). The forklift may not be used without safety equipment.

How to Consult the Manual

There is a table of contents at the beginning of the manual for ease of use. The manual is divided into chapters with specific topics. The name and title of the chapter are given at the top of each page The following is found at the bottom of each page: the type of manual, the identifying code, the language and the manual version.

Some general information is provided in this manual. Please only consider the information relevant for your specific forklift.

The following symbols have been used to highlight some parts of this manual.

⚠ DANGER

Failure to observe the instructions highlighted with this symbol may jeopardise safety.

⚠ CAUTION

Failure to observe the instructions highlighted with this symbol may cause damage to the forklift and, in some cases, result in warranty invalidity.

ENVIRONMENT NOTE

Failure to observe the instructions highlighted with this symbol may cause environmental damage.

NOTE

This symbol is used to provide additional information.

Date of edition and latest update of this manual

The publication date of these operating instructions is printed on the cover sheet.

The manufacturer makes continuous efforts to improve its industrial trucks, and therefore reserves the right to implement changes and to accept no claims concerning the information provided in this manual.

To receive technical assistance, please contact the service centre authorised by your closest manufacturer.

Copyright and trademark rights

These instructions must not be reproduced, translated or made accessible to third parties—including as excerpts—except with the express written approval of the manufacturer.

Delivery of the forklift and documentation

Ensure that the truck has all of the options requested and that it has been delivered with the following documentation:

- Original instructions
- Rules for the compliant use industrial vehicles;
- EC Declaration of Compliance;
- Warranty book.

If the forklift has been delivered with a traction battery and/or battery charger, ensure that such products conform to the order and that the relative user and maintenance manual are included, as well as the EC declaration for the battery charger.

If applied equipment, other equipment or devices are present, ensure that they conform to the order and that the relative use and maintenance manual and of the relative EC declaration (if provided by regulations in effect) are included.

All of the above documentation must be kept for the entire operative life of the forklift. In the event that the documentation is lost or damaged, contact the authorised sales network for copies of the original documentation.

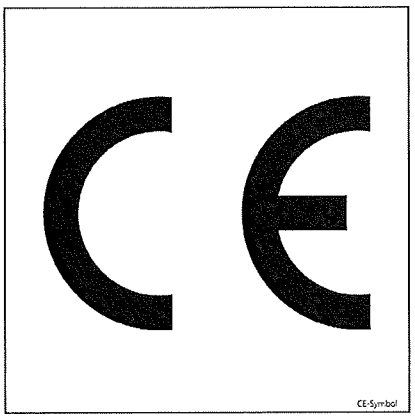
CE labelling

The manufacturer uses CE labelling to indicate that the truck complies with the standards and regulations valid at the time of marketing. This is confirmed by the issued EC declaration of conformity. The CE labelling is attached to the nameplate.

An independent structural change or addition to the truck can compromise safety, thus invalidating the EC declaration of conformity.

The EC declaration of conformity must be carefully stored and made available to the responsible authorities.

▷



CE-Symbol

EC declaration of conformity in accordance with Machinery Directive

Declaration

STILL GmbH
Berzeliusstraße 10
D-22113 Hamburg Germany

We declare that the

Industrial truck	according to these operating instructions
Model	according to these operating instructions

conforms to the latest version of the Machinery Directive 2006/42/EC.

Personnel authorised to compile the technical documents:

See EC compliance declaration

STILL GmbH

Technical service and spare parts

For scheduled maintenance and any repairs to the forklift, contact only the authorised service network.

The authorised service network has personnel trained by the manufacturer, original spare parts and the tools necessary to carry out maintenance and repairs.

Servicing by the authorised service network and the use of original spare parts maintain the technical characteristics of the forklift over time.

Only original spare parts provided by the manufacturer may be used for forklift maintenance and repairs. The use of non-original spare parts invalidates the warranty and renders the user responsible for any accidents due to the inappropriateness of the non-original parts.

Type of use

"Normal use conditions" of the forklift are understood as:

- lifting and/or transport of loads using forks with weight and load centre within the values provided (see Chapter 6 - Technical Data).
- transport and/or lifting on smooth, flat and compact surfaces;
- transport and/or lifting of stable loads uniformly distributed on the forks;
- transport and/or lifting with the load centre approximately on the forklift's median longitudinal plane.

⚠ DANGER

The forklift must not be used for other purposes.

Any other use renders the user solely responsible for injury/damage to persons and/or objects and voids the warranty.

The following scenarios are examples of incorrect use of the forklift truck:

- Transport on uneven (irregular or non-compact) surfaces
- loads that exceed the weight and/or load centre limits;
- transporting non-stable loads;
- transporting loads not equally distributed on the forks;
- transporting swinging loads;
- transporting loads whose load centre is considerably displaced with respect to the forklift's longitudinal median plane;
- transporting loads of dimensions such as to block the view of the operator when driving;
- transporting loads piled so high that they could fall onto the operator;
- travelling with a load over 300 mm off the ground;
- transporting and/or lifting people;
- pushing or pulling loads;
- moving upwards or downwards on a slope with the load facing downwards;
- turning at high speed;
- turning and/or moving sideways on slopes (upwards or downwards);
- colliding with stationary and/or mobile structures;

⚠ DANGER

Improper use of the forklift could cause it and/or at the load to overturn.

Working conditions

The forklift has been designed and built for internal transport.

Do not use beyond the limits of the climatic conditions indicated below:

- Maximum ambient temperature: +40°C
- Minimum ambient temperature: +5°C
- Altitude up to 2000 m
- Relative humidity between 30% and 95% (without condensation).

⚠ CAUTION

Do not use the forklift in dusty areas.

Using the forklift in environments with high concentrations of salty air or water could interfere with its proper operation and cause corrosion of metallic parts.

If the forklift must be used in conditions that exceed the limits indicated or, in any case, under extreme conditions (extreme weather, cold-storage rooms, presence of strong magnetic fields etc), appropriate equipment and/or use precautions are necessary. Contact the authorised sales network for more information.

⚠ DANGER

The forklift may not be used in environments in which there is a risk of explosion. It may not be used to handle explosive loads.

For forklifts that must operate in environments in which there is a risk of explosion or must handle explosive loads, appropriate equipment is necessary and must be accompanied by a specific EC Declaration of Compliance which replaces that of the standard forklift, and by the relevant User and maintenance manual.

Contact the authorised sales network for more information.

Modifications to Forklift

No modifications may be made to the forklift, otherwise the EC certificate and the warranty will become invalid, with the exception of:

- assembly of the options provided by the manufacturer
- assembly of applied equipment

for which it is necessary to refer exclusively to the authorised sales network

⚠ DANGER

If the forklift is equipped at the factory or later with devices that emit non-ionising radiation (such as radio transmitters, RFID players, data terminals, scanners, etc), the compatibility of such devices must be verified with the presence of operators using medical devices (such as heart pacemakers).

Applied equipment

To use equipment that has not been applied, please contact the authorised sales network, in order to:

- verify feasibility
- install the equipment
- add a label with the new residual capacity is
- provide documentation on the equipment (user and maintenance manual and EC certificate).

User obligations

Users must comply with applicable local legislation governing forklift use and maintenance.

Environmental considerations

Disposal of components and batteries

The truck is composed of different materials. If components or batteries need to be replaced and disposed of, they must be:

- disposed of,
- treated or
- recycled in accordance with regional and national regulations.

NOTE

The documentation provided by the battery manufacturer must be observed when disposing of batteries.

ENVIRONMENT NOTE

We recommend working with a waste management company for disposal purposes.

Packaging

During delivery of the truck, certain parts are packaged to provide protection during transport. This packaging must be removed completely prior to initial start-up.

ENVIRONMENT NOTE

The packaging material must be disposed of properly after delivery of the truck.

2

Safety

Safety guidelines

General Precautions

NOTE

Some safety regulations to be followed when using the forklift are listed below. These regulations integrate those in the manual "**Rules for approved use of industrial vehicles**".

General Safety Rules

- Only allow qualified, trained and authorized personnel to use the forklift.
- Do not install equipment on the forklift unless supplied or indicated by the manufacturer.
- Maintain the forklift in full working efficiency in order to limit any type of risk to the minimum.
- Do not use the truck with bonnets or doors open or with guards removed.
- The data plates found on the forklift must be kept in good condition and replaced if damaged.
- Carefully read and follow all of the safety indications found on the forklift.
- Make sure that the forklift has sufficient overhead clearance.
- Do not park the forklift in front of fire-fighting devices or fire escapes or anywhere that it blocks traffic.
- If the forklift shows signs of failure or breakage and there is reason to consider it unsafe, stop, park it, and notify the maintenance manager.
- Maintain appropriate distances from high voltage overhead cables. Comply with the safety distances established by the competent authorities.
- Never raise the load using just one fork.
- Place the load on the fork carriage or in such a way that the centre of gravity of the load is as close as possible to the fork carriage.
- The load must be placed on the fork arms so that the centre of gravity falls lengthwise on the mid point between the fork arms.
- Do not drive with loads off-centre laterally with respect to the forklift's median axis. Lack of compliance with this regulation can compromise forklift stability.
- Make sure that the surface on which the load rests is able to support its weight.
- Always use safety clothing compliant with current regulations and any personal protective equipment that may be applicable.
- Do not travel on loose or hilly ground or on steps.
- Do not drive with loads raised more than 300 mm from ground level.
- Do not turn or stack on slopes.
- Reduce speed on slopes.
- Do not overload the forklift beyond the capacity limits indicated on the capacity plates.
- Individuals under the influence of drugs and alcohol are not permitted to use the truck.
- The operator may not use an MP3 player or any electrical device that may distract their attention from the surrounding work environment.

Flooring requirements

The work floor must be even and free of holes or dips, which can be difficult to get around. Any steps must be equipped with ramps to prevent impacts with the wheels, which affect the entire structure of the truck.

⚠ CAUTION

Passing over cracks or damaged parts of the floor with the truck is prohibited. Dirt and any objects in the work path must be removed immediately.

Battery connection cables

⚠ CAUTION

Using sockets with NON-ORIGINAL battery connection cables can be dangerous (see purchase references in the parts catalogue)

Requirements for the traction-battery charging area

When the traction battery is being charged, the area must be sufficiently ventilated in order to download or eliminate the gases produced (EN 50272-3).

Safety Regulations Relative to Forklift Use

- The operator must familiarize himself with the forklift to be able to better describe any defects and assist maintenance personnel. The operator, trained and authorized to use the forklift, must be familiar with the controls and performances of the forklift.
- Any defect (squeaking, leaks, etc.) must be promptly reported because, if neglected, it could cause more serious failures/defects.
- Carry out the inspections indicated in the chapter on "Daily Inspections".

ENVIRONMENT NOTE

Report any oil and/or battery fluid leaks: they are dangerous and highly polluting.

⚠ CAUTION

If you notice a burning smell, stop the forklift and turn off the engine, then disconnect the battery.

Safety Regulations Relative to Operating Materials

Rules for handling and disposing of operating materials

ENVIRONMENT NOTE

Improper use and disposal of operating and cleaning materials can cause serious damage to the environment.

Always use and handle the operating materials in a suitable manner and follow the manufacturer's instructions for the product's use.

Keep the operating materials only in containers intended for this purpose and in a location that satisfies the requirements.

The operating materials may be flammable, so avoid contact with hot objects or open flames.

When topping up the operating materials, only clean containers should be used.

Follow the manufacturer's safety and disposal instructions regarding the operating and cleaning materials.

Do not disperse oils or other operating liquids! Any spilt liquid must be immediately collected and neutralised with a binding material (such as an oil binder) and then disposed of in accordance with current regulations.

Always comply with anti-pollution regulations!

Before carrying out work that involves lubrication, filter replacement or hydraulic equipment interventions, the area in question must be thoroughly cleaned.

The replaced parts must always be disposed of in accordance with the anti-pollution laws.

ENVIRONMENT NOTE

The incorrect or unlawful use of brake fluid is harmful to people's health and the environment.

Oils

- Do not allow to come into contact with the skin.
- Do not inhale oil vapors.
- Wear appropriate means of individual protection during forklift maintenance operations (gloves, goggles, etc.) to prevent the oil from coming into contact with your skin.

ENVIRONMENT NOTE

The used oils and relative filters contain substances that are hazardous to the environment and must be disposed of according to current regulations. We advise you to contact the authorised service network.

DANGER

The penetration in the skin of hydraulic oil that has leaked under pressure from the forklift's hydraulic system is dangerous. If this type of lesion should occur, contact a doctor immediately.

DANGER

Small high pressure jets of oil can penetrate the skin. Look for any leaks using a piece of cardboard.

Battery Acid

- Do not inhale the vapor: it is poisonous.
- Use adequate means of individual protection to prevent contact with the skin.
- Battery acid is corrosive: if it should come into contact with your skin, rinse abundantly with water.
- Explosive gas mixtures can form when charging the battery; therefore, the rooms in which the battery is charged must be in compliance with the specific regulations on the subject (e.g. EN 50272-3 etc.).
- DO NOT smoke or use open flames and lights within a 2 m radius from the charged battery and in the battery charging area.

NOTE

For greater information, consult the specific battery manual that comes with the battery.

ENVIRONMENT NOTE

The batteries contain substances that are hazardous to the environment. The replacement and disposal of the life-expired battery must be carried out as required by law. We advise you to contact the authorised service network that is equipped for eco-friendly disposal in accordance with current regulations.

Residual risks

Residual dangers, residual risks

Despite careful use and compliance with standards and regulations, the possibility of other risks occurring when using the truck cannot be entirely excluded.

The truck and all other system components comply with current safety requirements. Nevertheless, even when the truck is used for its proper purpose and all instructions are followed, some residual risks cannot be excluded.

Even outside the defined danger areas of the truck, residual risk cannot be excluded. Persons in this area around the truck must exercise a heightened degree of awareness, so that they can react immediately in the event of any malfunction, incident or breakdown etc.

⚠ WARNING

All persons that are in the vicinity of the truck must be instructed regarding the risks that arise through use of the truck.

In addition, we draw your attention to the Safety Guidelines in these operating instructions.

Risks can include:

- Escape of consumables due to leakages, rupture of lines and containers etc.
- Risk of accidents when driving on ramps or in conditions of poor visibility, etc.
- Falling, tripping etc. when moving the truck, especially in wet or icy conditions or when consumables are leaking.
- Fire and explosion risks due to batteries and electrical voltages.

- Human error resulting from failure to observe the safety guidelines.
- Unrepaired damage or defective and worn components.
- Insufficient maintenance and testing
- Use of incorrect consumables
- Maintenance intervals exceeded

The manufacturer shall not be held responsible for accidents involving the truck caused by the failure of the operating company to comply with these regulations either intentionally or due to negligence.

Stability

The stability of the truck has been tested in accordance with up-to-date technical regulations and is guaranteed if the truck is used correctly and in line with the intended purpose. These standards only take into account the static and dynamic tipping forces that can arise during use in accordance with the operating standards and intended purpose. In extreme cases there is a risk of exceeding the moment of tilt due to improper use or incorrect operation, which will affect stability.

Risks can include:

- loss of stability due to unstable or sliding loads etc.;
- turns at excessive speeds;
- moving with the load raised;
- moving with a load that is projecting to the side (e.g. side shift);
- turning and driving diagonally across slopes;
- driving on slopes with the load pointing downhill;
- oversized loads;
- swinging loads;
- steps or ramp edges.

Electromagnetic radiation

The limit values for electromagnetic emissions and for immunity relative to the forklift are those provided by the EN 12895 standard.

Non-ionised radiation

If the forklift is equipped at the factory or later with devices that emit non-ionising radiation (such as radio transmitters, RFID players, data terminals, scanners, etc), the compatibility of such devices must be verified with the presence of operators using medical devices (such as heart pacemakers).

Noise

Sound pressure level in driver's seat	L_{pAZ} < 70 dB (A)
Uncertainty factor	K_{pA}=4 dB (A)

The value is determined in a test cycle in accordance with Harmonised European Standard EN 12053 and declared according to EN ISO 4871 with weighted time percentages of the Transport, Lifting and Idling modes.

⚠ CAUTION

The value expressed above can be used to compare forklift trucks of the same category. This cannot be used to determine the noise level in workplaces (daily personal noise exposure). Noise values that are lower or higher than those indicated above can occur during actual truck use, for example following different operating modes, different environmental conditions and additional noise sources.

Vibrations

Vibrations to which the hands and arms are exposed

The following value is valid for all truck models:

- $\bar{a}_w < 2.5\ m/s^2$

NOTE

It is mandatory to specify the hand-arm vibrations, even where the values do not indicate any danger, as in this case.

CAUTION

The value expressed above can be used to compare forklift trucks of the same category. It cannot be used to determine the operator's daily exposure to vibrations during real operation of the truck; these vibrations depend on the conditions of use (floor conditions, method of use etc.) and therefore daily exposure must be calculated using data from the place of use.

Safety tests

Regular safety inspection of the truck ▷

Safety inspection based on time and extraordinary incidents

The operating company must ensure that the truck is checked at least once a year, or following noteworthy incidents.

As part of this inspection, a complete check of the technical condition of the truck must be performed with regard to accident safety. In addition, the truck must be thoroughly checked for damage that could potentially have been caused by improper use. A test log must be created. The results from the inspection must be retained until a further two inspections have been carried out.

The inspection date is indicated by an adhesive label on the truck.

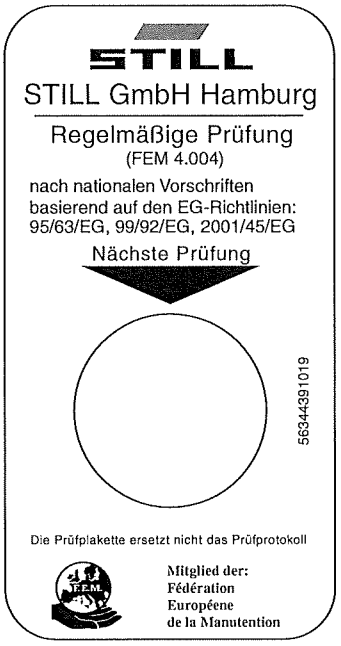
- Arrange for the service centre to perform periodic safety inspections on the truck.
- Observe guidelines for checks carried out on the truck in accordance with FEM 4.004.

The operator is responsible for ensuring any defects are remedied without delay.

- Contact your service centre.

NOTE

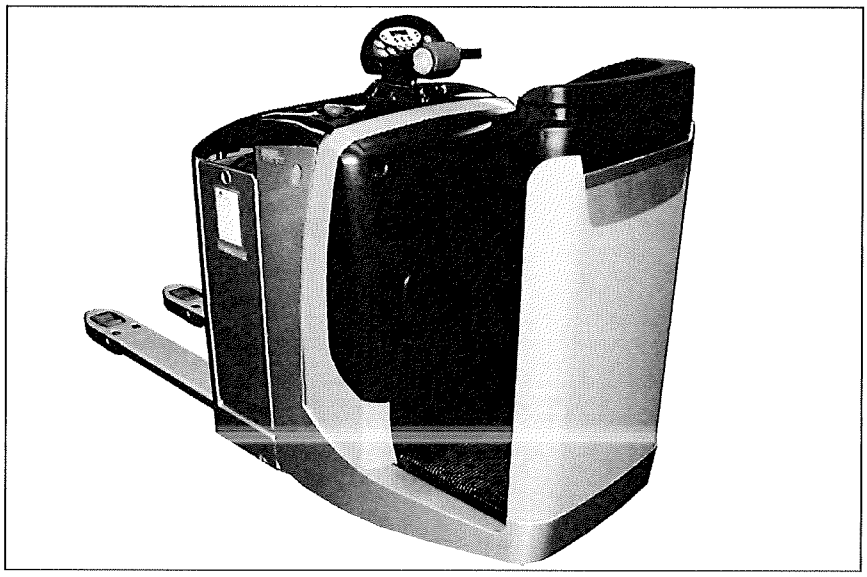
Observe the national regulations for your country!



3

Overview

Technical description



This robust stand-on electric pallet truck is ideal for loading and unloading lorries, and its high driving speed also makes it easy to transport over longer distances.

Chassis

- The EXU-S was designed for loading and unloading lorries and has excellent handling on slopes and in the confined spaces of lorries.
- The even chassis moves smoothly and never catches on protruding edges of ramps, whilst the platform's tapered sides protect the operator from colliding with side walls, even when working on the end two pallet rows.
- The platform is an integral part of the chassis and it has an enclosed, high-resistance structure providing extreme reliability, even on violent impacts on the dock.
- To optimise traction and braking, the centre-mounted drive unit is sprung and

features variable wheel pressure on the ground proportional to the load on the forks.
- For lateral stability, two exclusive STILL twin-wheels have been specially designed for the EXU-S for extreme reliability in the most arduous applications.

Steering

- Fully electric steering "with automatic return to neutral".
- In the centre angle range, any unintentional steering movements are damped by means of an electronic filter, guaranteeing precise steering.
- Automatic speed reduction during cornering combined with the stabilising wheels gives high lateral stability and reduces centrifugal force effects on the operator and the load.

Cockpit

- The new console "cockpit" contains all the functions for access, operation and control on the EXU-S.
- Due to the ergonomic layout of the controls, all functions can be operated with either hand.
- The following functions are integrated into the cockpit:
- Steering and speed control
- Raising/lowering of forks
- Hour meter connected to functions
- Battery discharge indicator with fork lift cut-out
- Operator profiles with two performance modes
- Performance mode selection by electronic key
- Service centre diagnosis for maintenance (by means of service codes)
- In addition, the cockpit provides a secure support for the operator whilst driving, so that he does not get tired or have to keep turning backwards and forwards.

Driver rides standing on "sprung" platform

- Depending on the application, the EXU-S is available with two platform types that are ergonomically suited to the operator's needs. These needs are very different from one type of job to another, so we have designed two different platforms.
- For typical loading/unloading applications where pallet transfer or long runs require the driver to spend a long time standing on the truck, the platform with side access will offer the greatest comfort and safety for driving in both directions.
- If the driver has to step in and out frequently for any peripheral work, (scanning products, manual handling etc.), then the platform with rear access is best suited.
- Either EXU-S platform type always provides great comfort due to an innovative, unique, fully sprung platform floor.
- The truck incorporates various storage facilities and a holder for rolls of stretch wrap integrated within the hood.
- A clipboard is available as standard.

Driving

- The asynchronous traction technology is powerful and economical. Travel power is 3.0 kW.
- The latest generation of asynchronous controller combined with the cockpit offers the possibility of selecting two drive modes by touching one of the two performance mode buttons:
- In "ECO" mode (Tortoise button), the controller supplies the travel motor with low current and it accelerates gradually. This results in 15 % more battery economy, but the same productivity as other machines of this type.
- In "BOOST" mode (Hare button), the controller drives the travel motor to provide very high torque and acceleration, even when fully loaded. While economy remains unchanged, productivity increases by around 25 %.

- With each mode, the drive characteristics can be adjusted (speed, acceleration, braking) to precisely match the application or driver preference.
- Thanks to the controller's speed control system, the EXU-S will start smoothly and accelerate to maximum travel speed, regardless of the load on the forks.
- The service brake is activated by releasing the butterfly valves or by reversing the direction of travel.
- To avoid any unintentional movements whilst on a ramp, the (electromagnetic) parking brake is activated once the butterfly valves are in neutral position or the operator leaves the platform.

Hydraulic system

- A pump unit comprising a compact, 2.2-kW pump motor with built-in oil tank, solenoid valve and safety valve controlling electrical end-of-travel lift cut-out.
- This hydraulic system offers very short lift times, even with a full load, thereby providing high productivity for loading and unloading lorries with the EXU-S.

Brake circuit

- There are two independent brake circuits.
- Progressive service brake with energy recovery activated by releasing or reversing the butterfly valves.
- Emergency brake with electromagnetic disc activated by a button in the cockpit.
- Driving is only possible when the operator is on the platform, which acts as a contact switch.

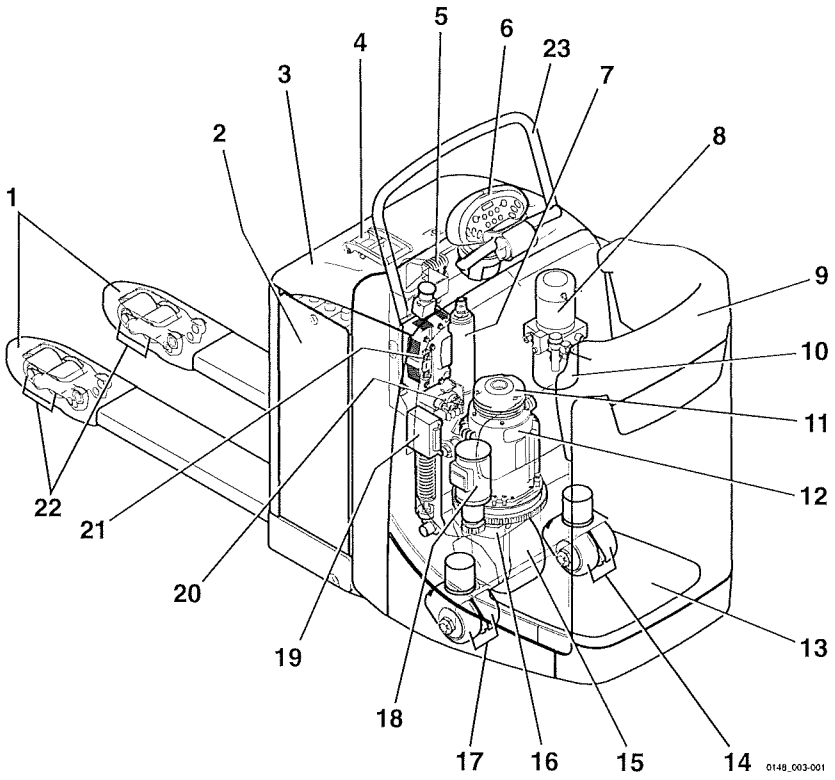
Battery

- The battery is easily accessible which makes it easier to recharge or carry out maintenance work (for example, "filling the water up"). The batteries can be replaced

vertically with a sling, or from the side using rollers, if used for 2 or 3 work shifts.

- Two different battery compartments allow batteries with 450 Ah or 600 Ah capacities to be used.

Overviews

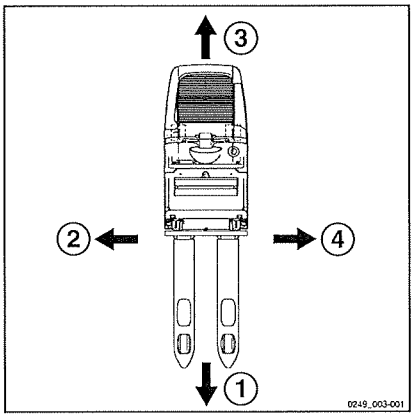


1	Forks	11	Brake
2	Battery compartment	12	Drive motor
3	Battery cover	13	Platform (interlock zone)
4	Battery lock	14	Stabiliser
5	Battery connector	15	Traction wheel
6	Cockpit	16	Gearbox
7	Lift cylinder	17	Stabiliser
8	Pump motor	18	Steering motor
9	Backrest	19	Steering controller
10	Hydraulic oil tank	20	Fuse mount

21 Drive controller
22 Load wheel
23 Mounting bracket for accessories (option)

Definition of directions

The references forwards (1), backwards (3), right (2) and left (4) when describing the position are relative to the driver's seat. The load is at the front.

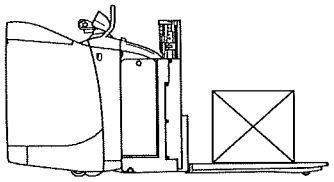


Sample graphics

This documentation explains the (usually sequential) sequence of certain functions or operations. Schematic diagrams of a truck are used to illustrate these sequences.

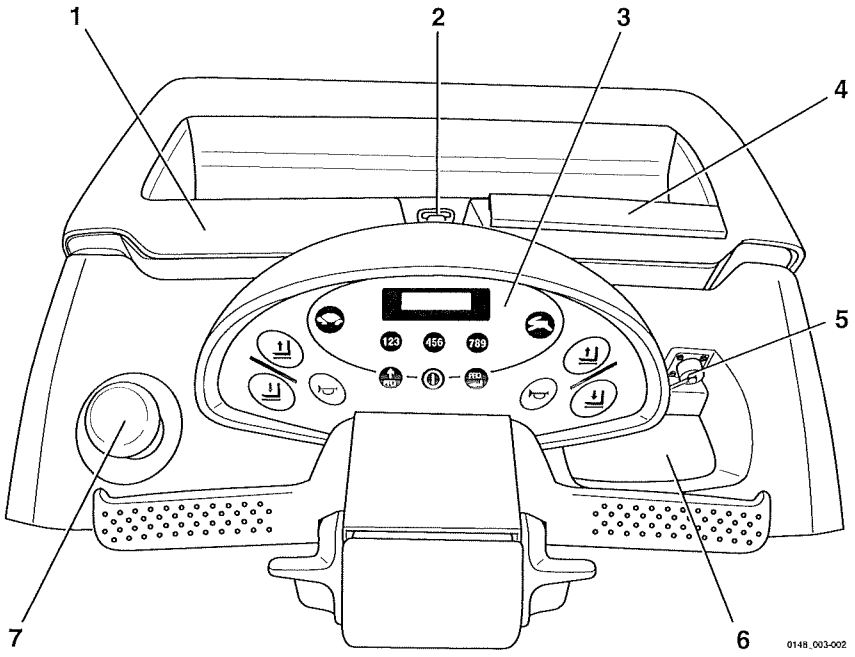
NOTE

These schematic diagrams are not representative of the constructed state of the documented truck. They serve only to illustrate the sequences.



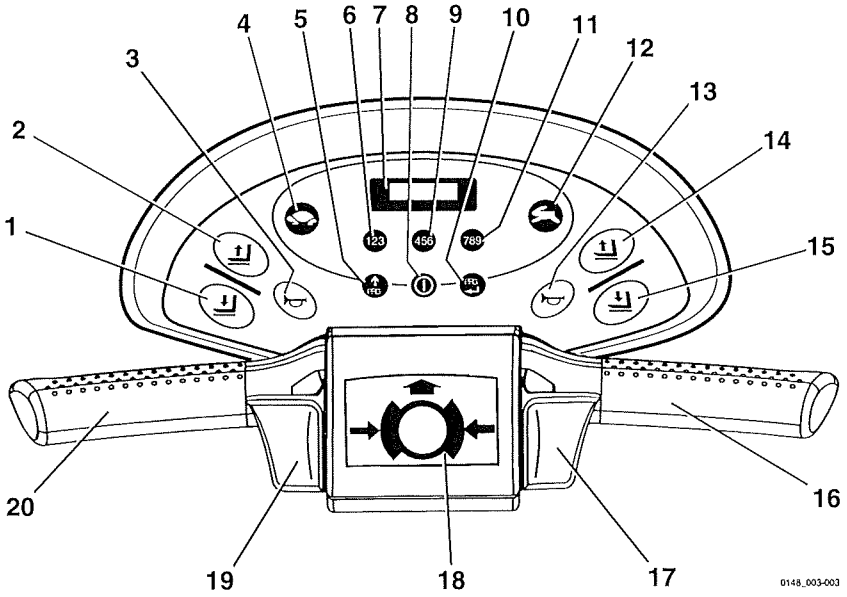
Control and display elements

View of the control elements



1 Battery cover
2 Battery compartment lock
3 Cockpit
4 Document holder
5 Diagnostic connector
6 Glove compartment
7 Emergency isolator button

Cockpit



1	Lower button	11	Numbers 789 button
2	Lift button	12	Hare button
3	Horn button	13	Horn button
4	Tortoise button	14	Lift button
5	PRG ↑ (scrolling) button	15	Lower button
6	Numbers 123 button	16	Handle
7	Display	17	Travel control
8	I (ON) button	18	Emergency brake switch
9	Numbers 456 button	19	Travel control
10	PRG ↵ input (enter) button	20	Handle

Dead man's zone

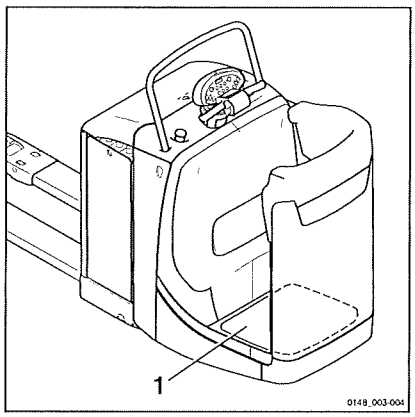
⚠ CAUTION

The pallet truck has a foot interlock zone which can cause the truck to be braked.

Method of operation

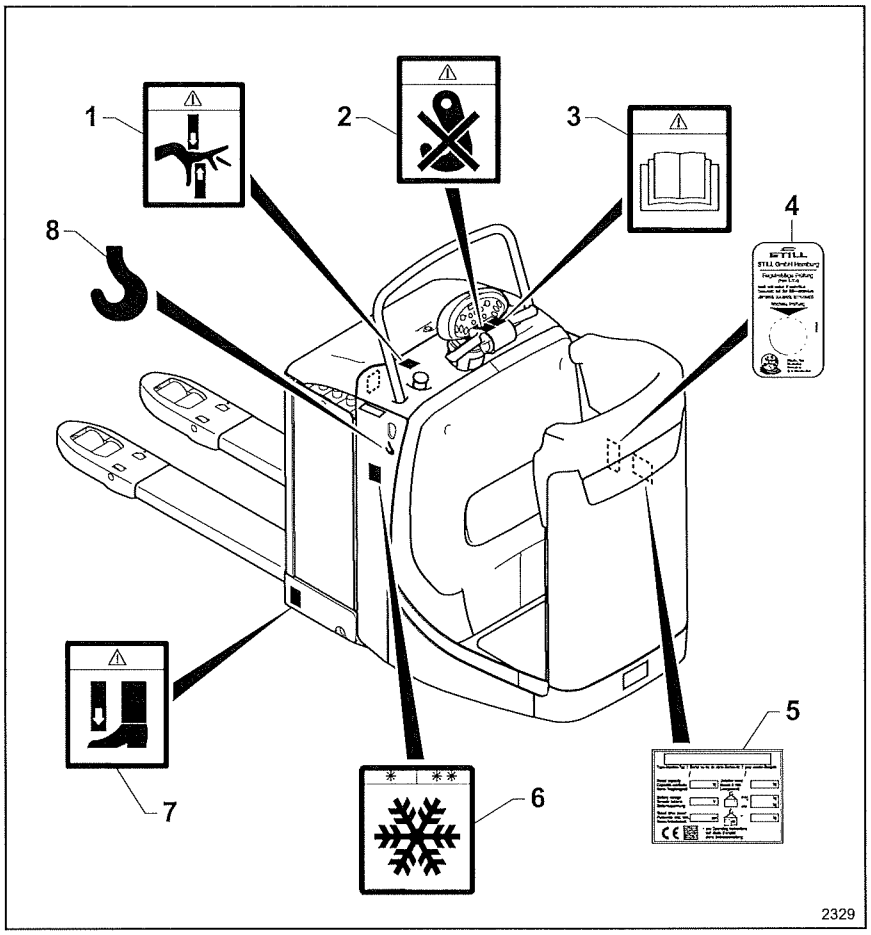
The driver must stand with one foot on the foot interlock zone (1) of the driver's platform. The feet must stay in the area of this platform.

The pallet truck is braked as soon as the foot interlock is no longer activated (the driver leaves the platform).



Markings

Location of designation plates and labels



1	"Danger of crushing hands" label	5	Nominal value designation plate
2	"No hooking" label	6	"Cold store" label (on cold store version only)
3	"Operating and maintenance manual" label	7	"Danger of crushing feet" label
4	Annual testing label (Germany only)	8	"Hook" symbol

Description of labels

(1) The label indicates the danger and risk of crushing hands area.

(2) This label indicates that you must not lift the truck by hooking onto the cockpit

(3) This label indicates that you should consult the operating and maintenance manual before using the truck and before carrying out any maintenance work.

(4) This label is only present on trucks sold in Germany. The label indicates the date of the truck's periodic safety inspection.

(5) This designation plate gives the identification data for the truck

(6) This symbol, where present, indicates that the truck is set up for use in cold-storage (option).

(7) The label indicates the danger and risk of crushing feet area.

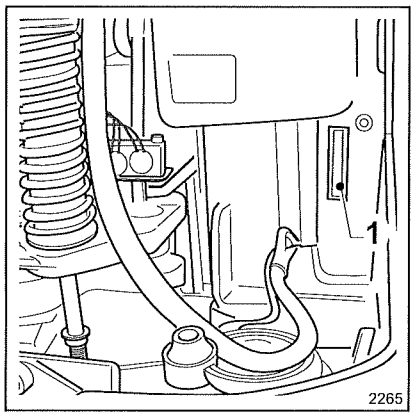
(8) This label indicates where to attach the truck's lifting hook.

Warning plates

Always observe the indications on the warning plates.

Chassis frame labelling

The truck's serial number is marked on the chassis frame (1).

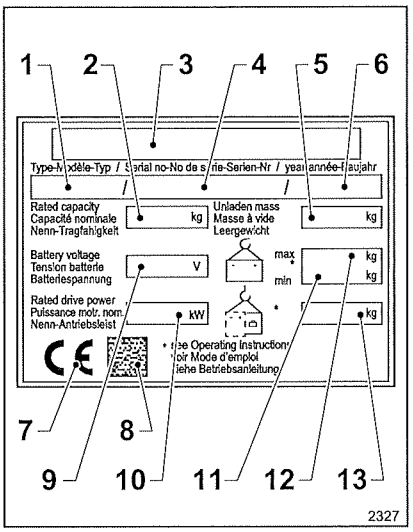


Data plate

NOTE

Please indicate the serial number for all technical enquiries.

▷

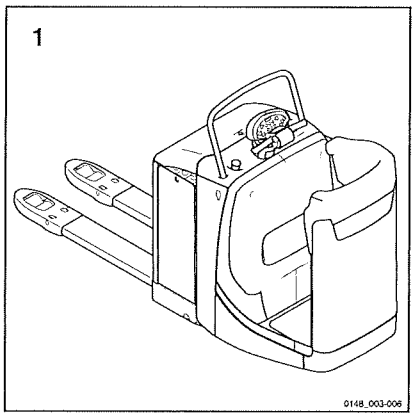


1 Model
2 Rated capacity in kg
3 Manufacturer
4 Serial no.
5 Unladen weight (without battery) in kg
6 Year of manufacture
7 EC conformity symbol
8 QR code
9 Battery voltage V
10 Nominal power rating in kW
11 Minimum battery weight
12 Maximum battery weight
13 Additional weight (ballast) in kg

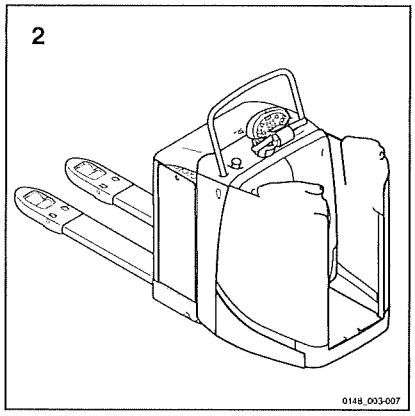
Options

Models

1 Platform, access on left side ▷



2 Platform, access at rear ▷



Options

- Single load wheels
- Continuous operation hour meter (with own battery in the tiller head)
- Battery charge indicator (for gel battery)
- Battery charge indicator (for gel battery) with service interval indicator
- Battery charge indicator (for lead battery) with service interval indicator

- Cold store version
- Document holder
- Preparation for data input terminal
- FleetManager
- Tray 71
- Various fork gauges and fork lengths
- Various load rack sizes
- Various types of tyre for the drive wheel
- Slow speed with forks lowered
- Battery electrolyte level indicator LED

Accessories

- Master code (factory setting "1 2 3 4")
- Key for battery compartment (2 items)

Options

4

Application

Authorised and safe use

Intended use of the trucks

⚠ CAUTION

This machine is intended for the transport of loads packed on pallets or in industrial containers designed for this purpose, as well as for placing pallets into and removing pallets from stock.

The dimensions and capacity of the pallets or containers must be adapted to the load being transported to ensure stability.

The table of characteristics and performance attached to this user manual gives you some of the information you need to check that the equipment is suitable for the work being carried out.

Any specific usage must be authorised by the site manager; an analysis of the potential risks associated with this usage will enable him to put in place any necessary additional safety measures.

Safety instructions relating to use of the truck

Behaviour when driving

The operator must obey the same rules within the plant as on the road. The operator must drive at a speed appropriate for the driving conditions. For example, the operator should drive slowly around corners, when entering and travelling through narrow passageways, when driving through swing doors, at blind spots, or on uneven surfaces. The operator must always maintain a safe braking distance from vehicles and persons in front of him and must always have the truck under control. The operator must avoid sudden stops, making fast U-turns and overtaking other vehicles in potentially dangerous areas or areas with poor visibility.

⚠ WARNING

Driving the truck while sitting down is prohibited.

Please remember the following:

- Drive the truck as described in the "Operator positions" section.
- The truck must not be used as a stepladder.

- The truck has not been designed to transport anyone other than the operator and must not be used for this purpose.
- The operator must always stay within the truck clearance.
- Stay in the safety area (working area defined by the manufacturer).

NOTE

Using a telephone or radio in the truck is permitted, but avoid using these devices when driving as they may distract you.

People in the danger area

Before starting the truck and while you are working, ensure that no one is in the danger area. If people are in danger, warn them well in advance. Stop working with the truck immediately if the people do not leave the danger area despite the warnings.

⚠ DANGER

Risk of injury! There is a risk of physical injury inside the danger area. Danger of death from falling loads!

Do not stand on the forks!

Standing or walking under the forks is strictly forbidden, even when they are not loaded!

Danger area

The danger area is the area in which people are in danger from the forklift truck movements, from its work equipment and from its load lifting devices (e.g. accessories) or from the load. The danger area also includes areas in which a load could fall or in which work equipment could lower or fall.

Traffic route conditions

The surface of traffic routes must be sufficiently level, clean and clear of objects. Drainage channels, railway crossings and other similar obstacles must be levelled and, if necessary, fitted with ramps so that the truck can cross without jolting.

There must be sufficient distance between the highest part of the truck or the load and the surrounding fixed installations. The height depends on the lift height and the dimensions of the load. Refer to the technical characteristics.

Regulations regarding the traffic routes and the manoeuvring areas

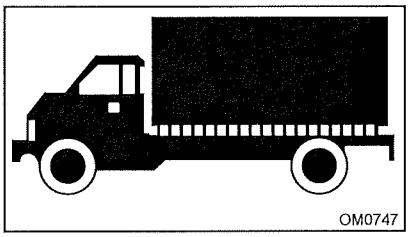
Only traffic routes authorised by the operator or his agent may be used. Traffic routes must be free of obstacles. Loads may only be unloaded and stored in places designed for this purpose. The operator or his agent must ensure that no unauthorised person approaches the working area.

Hazards

Hazards on the traffic routes must be signalled by standard road signs or possibly by additional warning notices.

Truck transport and lifting

Transporting the truck



The forklift is normally transported by road and rail. If the forklift's dimensions exceed the max. clearance size allowed, it is transported disassembled. The sales network is in charge of the disassembly and reassembly operations. The forklift must be secured to the transport means during transport using appropriate restraint systems. Block the wheels with wedges to prevent even the slightest movement.

Transport

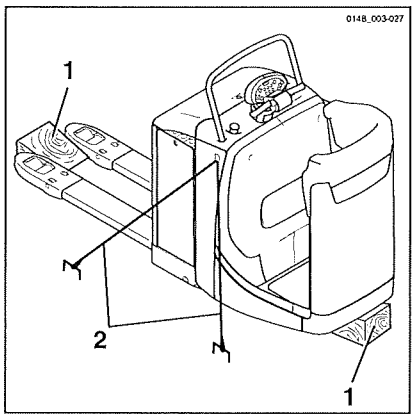
- Shut down the truck with the key switch.
- Disconnect the battery connector.

Chocking the truck

- Secure the truck against rolling and sliding with chocks (1). ▷

Lashing down the truck

- Attach the lashing ropes (2) at the frame.



Climatic Conditions for Transport and Storage

The forklift must be protected from atmospheric agents during transport and storage.

Breaking-In

This type of forklift does not require special breaking-in operations.

Checks and actions prior to use

Visual checks

⚠ WARNING

Damage or other faults on the truck or attachments (special equipment) can result in accidents.

If damage or other faults are noticed on the truck or attachments (special equipment) during the following checks, do not use the truck until it has been properly repaired. Do not remove or disable the safety systems and switches. Do not change the pre-set values.

⚠ CAUTION

Only use the truck if all of the covers are fitted correctly and the covers and doors are closed correctly.

⚠ CAUTION

Perform checks on a flat surface. Make sure that there are no people or objects in the test area in front of and/or behind the truck.

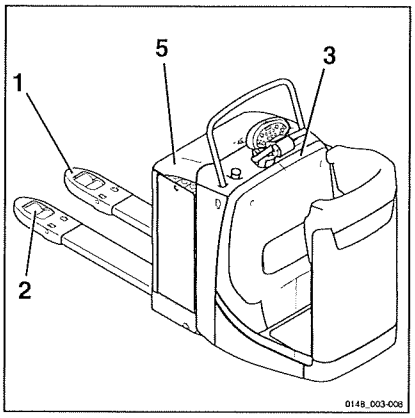
⚠ CAUTION

Drive very slowly during the operational tests.

Ensure that the vehicle is in good working condition prior to start-up. These checks supplement and do not replace the scheduled maintenance operations.

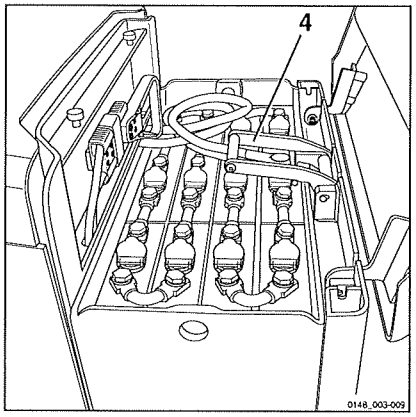
- No visible damage (e.g. deformation, cracks, heavy wear, hairline cracks) to forks (1) and other lifting devices.
- Inspect the wheels and load wheels (2) for strings, fibres and any other objects that could hinder their free rotation.

▷



- The latch (4) must be closed.

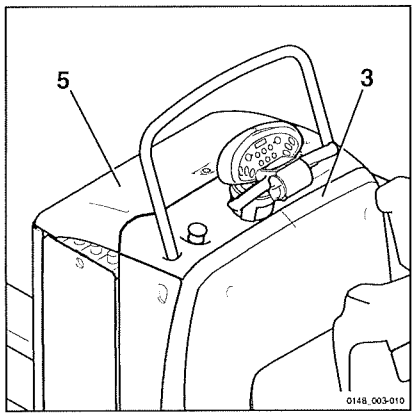
▷



Checks and actions prior to use

– The battery cover (5) must be closed.

– The cover (3) must be fitted.



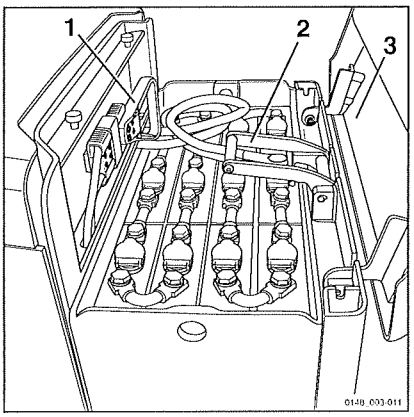
- Check that the warning and safety labels are complete and intact; see ⇒ Chapter "Location of designation plates and labels", P. 3-31.
- Replace any damaged or missing labels according to the overview of the location of plates and labels
- Attachments (optional) must be properly secured and operated in accordance with their user manual
- Inspect hydraulic pipes and hoses for visible damage and leakage. Replace damaged hoses
- Check that there are NO oil leakages in the area under the truck
- Visually check the uncovered sections of hydraulic hoses and pipes to ensure that they are in good condition and to detect any oil leakages
- Check that there are no objects (wires of various types, nails, screws, pieces of tape etc.) impeding the operation of the wheels and rollers. The wheels and the load rollers must roll freely
- The wheels must not show any sign of damage or heavy wear. They must be correctly mounted
- Check that the battery hood is fully and properly closed
- Check that all of the hoods and protective guards are present and check that they are correctly mounted
- There must be no objects on the truck that may limit visibility
- Check that NO stickers are missing or damaged. Replace damaged or missing stickers in compliance with the marking position table
- Visually check that the forks or other load-carrying equipment show NO obvious damage (e.g. bends, cracks, significant wear)
- Check that the battery male connector and female connector are fully intact and in good

condition. Check that they are working correctly

- Check that the start/stop key works correctly
- Check that the horn works correctly
- Check that the buttons and the control throttles on the tiller are working correctly
- One at a time, push the buttons and then release them. Check that the buttons return automatically to their initial positions. The buttons must not remain activated or stuck
- Turn the drive control throttle and then release it. Check that the throttle automatically returns to the initial position when it is released. The throttle must not remain activated or locked
- Test that the truck brakes and stops when the throttle is released during travel
- Check that the emergency stop button is operating correctly. Carry out the test when travelling towards the forks
- Check that the brake is operating correctly
- Test that the electromagnetic brake works effectively
- Check that the battery harness is in good condition
- Check and test the battery electrolyte level and density as indicated in the battery instructions
- The operator must be qualified to drive the truck. The operator must be able to reach the controls and operate them (especially the anti-crush protective device). Do not obstruct access to the controls

Connecting the battery plug (charging station)

- Disconnect the battery socket (1) from the charging station and insert into the plug on the truck.
- The latch (2) must be closed.
- Close the battery cover (3).



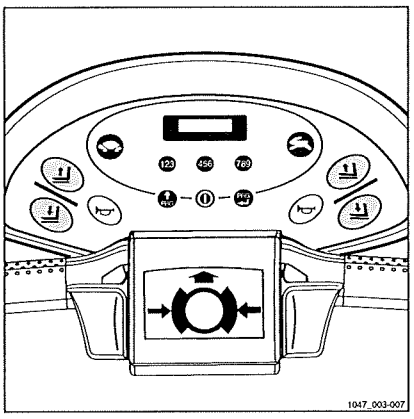
Start-up

Turning on the truck

NOTE

The pallet truck is equipped with a digital control and is taken into operation by entering a driver code. The driver code consists of four figures and is assigned by the pool manager. It is used for normal use and operation of the truck by the driver and operator. The knowledge of the code is like a key and should not be misused. Do not tell third persons your driver code except if expressly ordered to do so.

- It is possible to send other codes; see chapter: ⇒ Chapter "Digicode control", P. 4-71.



Entering the user code

- Press the [I] key (on) (6). The display (4) shows the message Code (8). This message switches off after approximately one second. You can then enter the 4-digit user code. The digits are entered in the same way as on a mobile phone.
- Press keys (1), (2) or (3) several times until the desired digit is shown;

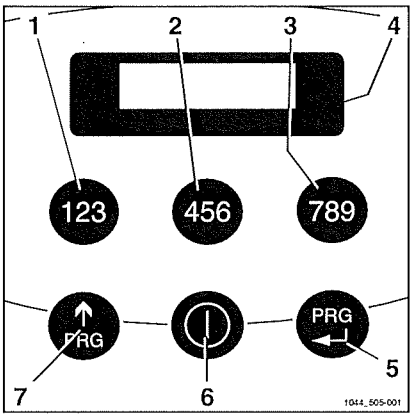
for digits:

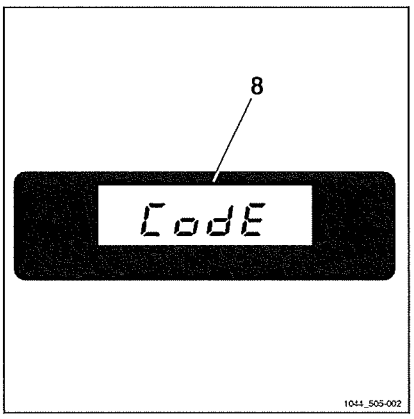
- 1-2-3, press key [123] (1),
- 4-5-6, press key [456] (2),
- 7-8-9, press key [789] (3).

After entering a digit, the display moves to the next position. The display shows a flashing line (9) (the illustration shows an example).

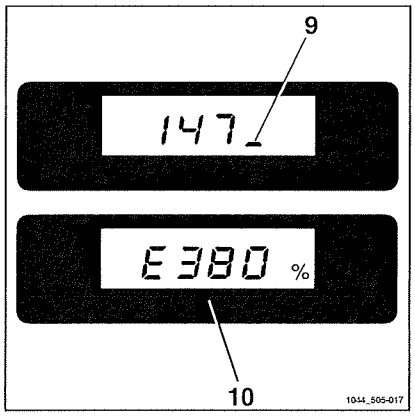
NOTE

If the digit entered is wrong, it is possible to cancel it by depressing the [PRG ↑] key (scroll) (7).





- The digit will be deleted and the flashing line (9) will reappear. It is now possible to enter digits again. ▷
- After the fourth digit has been entered, confirm this code by pressing PRG ↵ (5). The name of the truck manufacturer is displayed. It is now possible to activate the truck.



NOTE

If an error code is displayed, e.g. E380 *(10), stop the truck by pressing the key (6) and carry out the activation procedure again.*

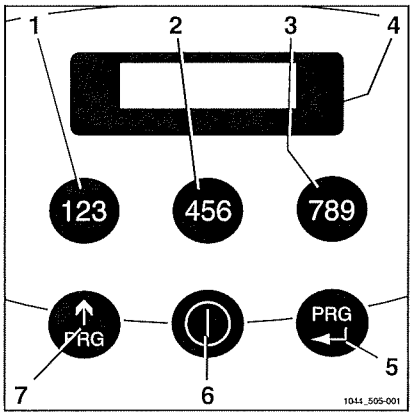
NOTE

It is possible to show the following displays by repeatedly depressing the PRG ↑ *key (scroll) (7).*

- Hour meter ⇒ Chapter "Hour meter", P. 4-49
- Battery charging ⇒ Chapter "Battery charge", P. 4-49
- Servicing interval management ⇒ Chapter "Display of time to next servicing", P. 4-50

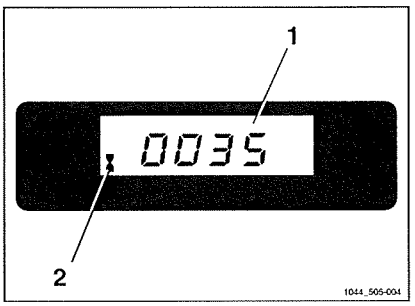
Starting with FleetManager (optional)

- Push the I key (START) (6). Do not pay attention to the Code alert that appears in the display ▷
- Start the truck directly via the FleetManager keyboard or reading device, depending on the version.



Hour meter

The operating hours (1) appear approx. 4 to 5 seconds after starting the truck, accompanied by the hourglass symbol (2). The total hours are indicated from when the truck is first commissioned. The hour meter totals during travel. Perform maintenance work according to the operating hours in accordance with the maintenance schedule.



NOTE

The meter goes to 9999.

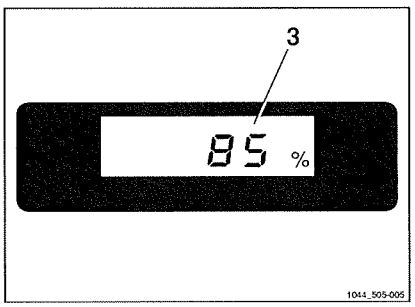
Battery charge

⚠ WARNING

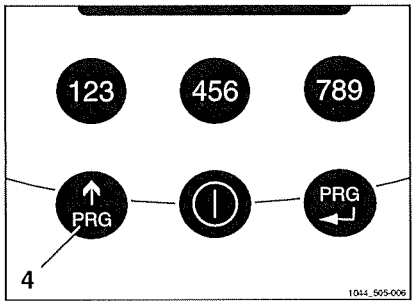
Do not place any metallic objects or tools on the battery. No naked lights, no smoking permitted! The battery could be damaged, shorted or an explosion could occur.

⚠ WARNING

The electrolyte (diluted sulphuric acid) is poisonous and caustic! Follow the safety precautions for handling battery electrolyte.



– Press the PRG ↑ (scrolling) button (4) to display the available battery charge (3).



NOTE

After the battery is connected, the correct state of charge appears only after about 1 minute.

The charge is shown in per cent and changes in steps of 5 %.

NOTE

The factory setting of the battery charge indicator is for a lead battery. The indicator can also be set to a gel battery. Please contact your Service.

- If a residual charge of 20% is reached, the battery must be charged soon.
- The driving speed is reduced.

⚠ CAUTION

Deep discharging damages the battery.

The battery must immediately be charged as deep discharging begins.

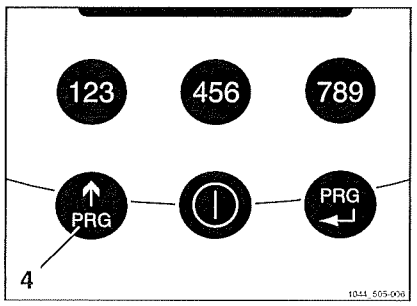
Display of time to next servicing

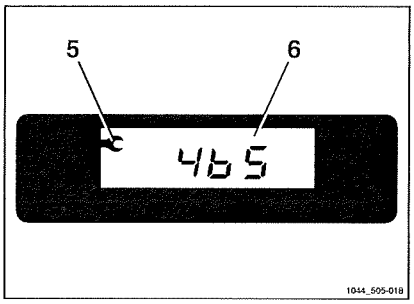
- Press the PRG ↑ (scrolling) button (4) to display the time remaining until the next scheduled servicing. The time is shown in hours (6) and with the symbol of a spanner (5). Have the maintenance carried out according to service hours specified in the maintenance schedule.

NOTE

The display of the time to the next servicing can be set by your Service. Please contact your Service in this regard.

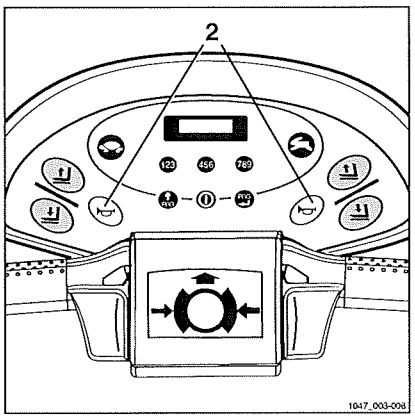
- The display can show subsequent information ⇒ Chapter "Error code", P. 4-79.





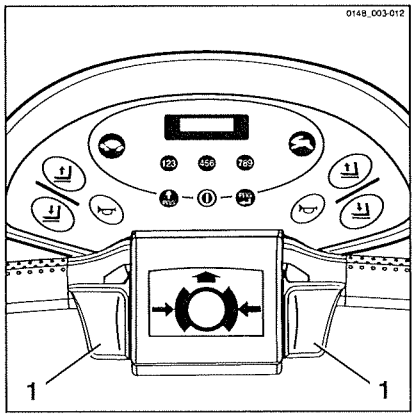
Operating the horn

- Press the horn button (2) on the cockpit.

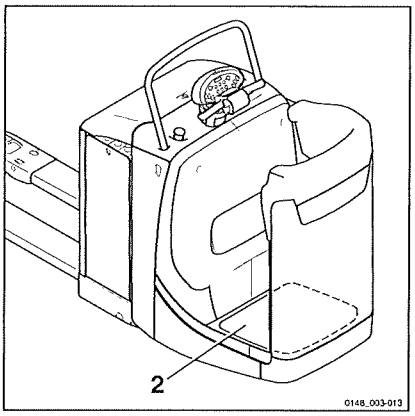


Check of brake and presence area

- Start the truck slowly with the throttle (1).

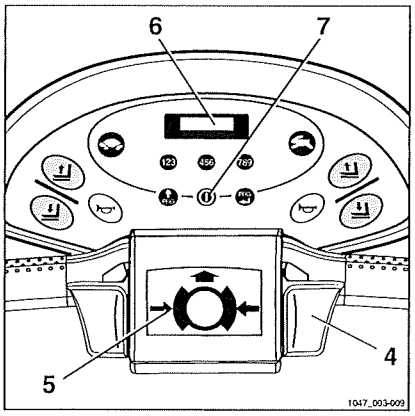


– Step off the presence area (2) while leaving the throttle actuated. ▷



Checking the emergency brake button ▷

– Drive the truck slowly using the traction control (4).
– Press the emergency brake button (5). The truck will be braked and remains stationary.
– The `StoP` signal appears on the display (6).



NOTE

Press the button [I] "ON" "7" to resume work. After `StoP` disappears from the display, the truck is operational again.

⚠ CAUTION

Only use this brake system in the event of an emergency!

Driving

Driving safety rules

Behaviour when driving

The driver must drive the truck within the plant according to regulations for driving on public roads. The speed of the truck must be appropriate for local conditions. For example, speed must be reduced when cornering, in narrow passages, when driving through swing doors, in areas with reduced visibility and/or on uneven ground. A safe distance must always be maintained from people and any vehicles in front. Sudden braking, fast turns, overtaking in hazardous areas or areas with poor visibility must be avoided.

Do not sit on the truck when driving.

EXU-S 22/24 trucks are designed for operation in ride-on mode, therefore:

- Do not operate the truck when sitting.
- Never use the truck as a climbing aid.
- Never use the truck to transport persons.
- Do not lean out. Always stay within the truck profile.

The use of a mobile phone or radiophone when operating the vehicle is permitted. However, do not use these devices while driving, as this will reduce your concentration.

First practice driving in a clear area or on a free roadway.

Visibility when driving

The driver must look in the direction of travel and have a clear view of the route travelled on. He must always ensure that the way is clear, particularly when reversing. When transporting goods that obstruct visibility, the truck must be driven with the load trailing. If this is not possible, a guide must walk ahead of the truck. In this case, the truck may only be driven at walking pace and with the utmost caution. The truck must be stopped immediately when eye contact with the guide is lost.

Before driving

Persons in the danger area

Before and during operation of the truck, the driver must ensure that no person is standing in the danger area of the truck. A warning signal must be given in the event of danger to persons. If these persons refuse to leave the danger area despite the warning, the driver must stop working with the truck immediately.



⚠ WARNING

Risk of injury!! Within of the danger zone there is a risk of being injured.

Do not step on the forks.



⚠ DANGER

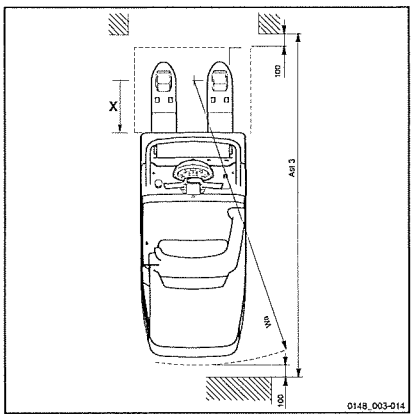
Risk of death through falling loads!

It is expressly forbidden to stand under the raised forks or to walk by beneath them, even if there is no load on the forks.

Danger zone

The danger zone is the area in which persons are endangered through the movement of the truck, its implements, load-lifting devices (eg attachments) or the carried load. This also includes the area within the range of falling loads or a descending implement.

Dimensions of roads and working aisles



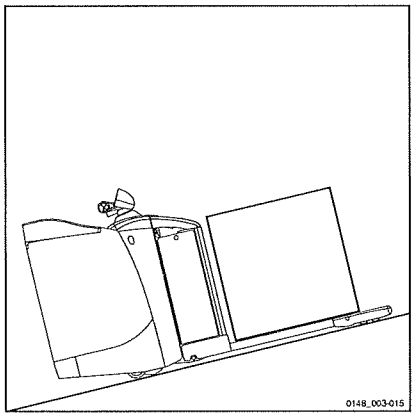
The dimensions specified in the description section apply under established conditions and should ensure safe manoeuvring. Check in each case whether a wider aisle is required when, for example, the load dimensions deviate.

The required aisle widths (Ast) depend on the dimensions of the load and are calculated according to the following formula: Ast = Wa - X + length of pallet + 200 mm.

Observe your local and national regulations.

Please see ⇒ Chapter "VDI data sheet for standard EXU-S 22", P. 6-101.

The truck may only be used on roadways without any tight curves, excessive slopes or passages that are too narrow or too low.



The following inclines or slopes may be driven on with the truck:

The slopes must not exceed the above gradients and should have a rough surface.

Smooth passages at the top and lower end should prevent the load from touching the ground or damage to the truck.

Condition of roadways

Roadways must be solid enough, free of debris and fallen objects. Drainage ducts, tracks and the like must be level and, if necessary, provided with ramps so that they can be crossed without jolts, if possible.

There must be sufficient clearance between the highest parts of the truck or the load and the surrounding fixtures. The height depends on the lift height and the load dimension. Check the datasheet for the truck.

Rules for roadways and working area

Only roadways made available by the operator or his representative may be driven on. The roadways must be free of obstacles. The load may only be stacked and stored in the appropriate places. The operator and his

representative must ensure that unauthorized persons keep away from the working area.

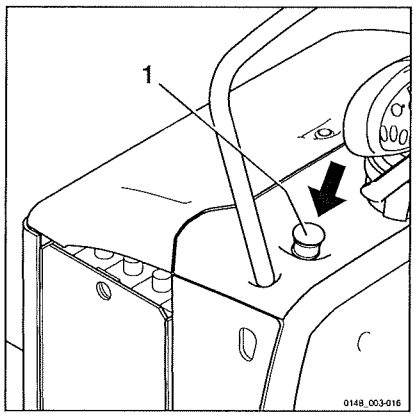
Hazard areas

Hazardous areas on roadways must be marked with the usual traffic signs or, if necessary, by additional warning signs.

Emergency stop procedure

In an emergency, all truck functions can be cut off.

– To do so, depress the emergency isolator button (1). The truck comes to a stop. ▷



⚠ CAUTION

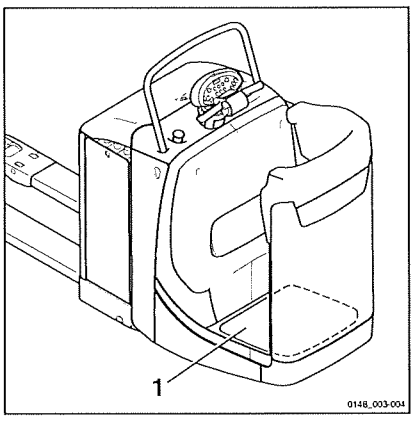
This safety feature may only be used in an emergency.

NOTE

After pulling up the emergency isolator button re-enter the driver's code to resume work.

Driving in rider mode

- Step on the platform with both feet and step on the presence area (1). ▷



NOTE

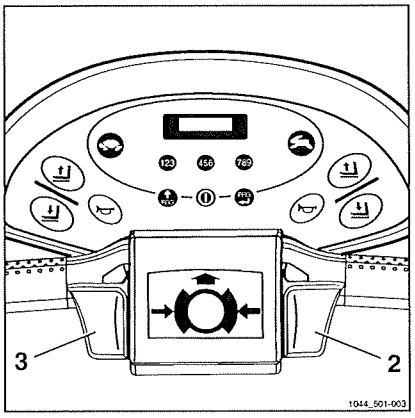
The pallet truck is turned on.

⚠ WARNING

There should be no objects lying on the floor of the driver's platform, whose weight could activate the dead man's switch under the mat.

Driving forward

- Depress the top part of the throttle (2) or (3) with the thumb. ▷
- Truck speed is increased by depressing the travel control further.
- When the travel control is released, the truck is braked electrically.



Reversing

- Depress the lower part of the throttle (2) or (3) with the thumb.
- Truck speed is increased by depressing the travel control further.
- When the travel control is released, the truck is braked electrically.

Reversing direction

- Allow the travel control to return to its neutral position.
- Operate the travel control progressively in the opposite direction of travel until the desired speed is reached.

Selecting the driving mode

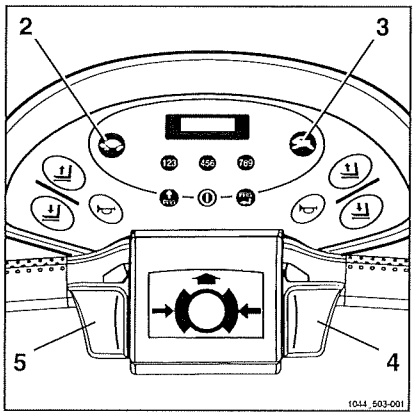
NOTE

The pallet truck has 2 selectable driving modes.

- For normal travel, push the Hare button (3).
- To drive with gentle accelerations, push the Tortoise button (2).

NOTE

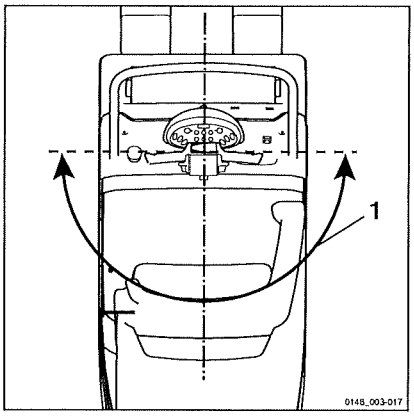
The travel control (4) and (5) can be operated with both the left and the right hand. Always operate the travel control slowly to avoid jerky starting, braking or reversing.



Steering

The truck is steered by turning the cockpit in range (1).

The turning radius (Wa) depends on the fork length and the battery. (See ⇒ Chapter "VDI data sheet for standard EXU-S 22", P. 6-101)



Brakes

⚠ WARNING

The surface finish influences the driving and braking behaviour of the truck.

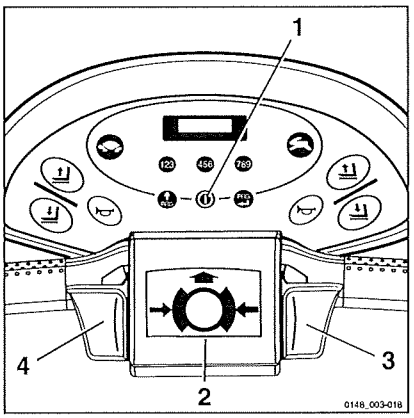
Therefore take this into account when driving and braking (eg: wet floor).

Soft braking

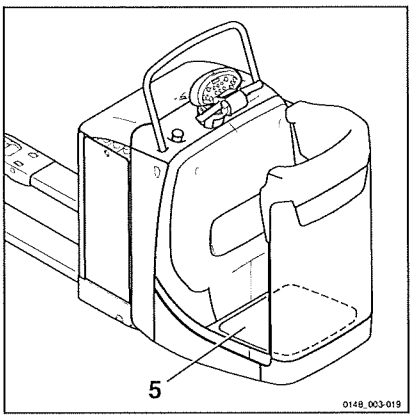
NOTE

The truck is braked electrically.

- Release throttle (3) or (4) when driving. ▷



- Leave the presence area (5). ▷



Average braking

- Operate the throttle (3) or (4) in the opposite direction.

Emergency braking

⚠ CAUTION

This type of braking may only be used in an emergency.

- Press the emergency brake button (2). The pallet truck stops.
- `StoP` appears in the display.

NOTE

Press the button ⏻ "ON" (1) to resume work. After `StoP` *disappears from the display, the truck is operational again.*

The truck is operational again.

Parking brake

- Release traction control (3) or (4).

The truck is braked by the electromagnetic brake.

Automatic shutdown

NOTE

If you leave the truck with the control turned on, it will be shut off automatically after approx. 15 minutes. The truck can only be operated again after the driver code is entered. The delay time is adjustable. Please contact your Service Engineer in this regard.

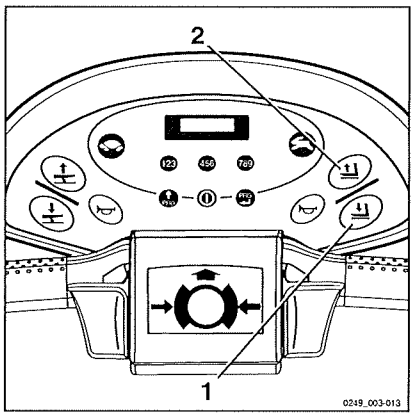
Lifting

Operating the main lift

Elevating the forks

– Press the lift button (2). The forks are raised. Their movement can be stopped at any time by releasing the button.

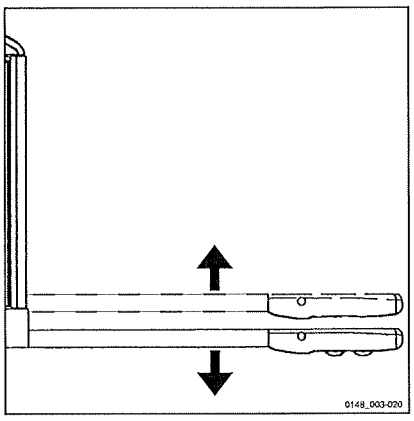
▷



Lowering the forks

- Press the lower button (1). The forks are lowered. Their movement can be stopped at any time by releasing the button.

▷



Load placement

Safety regulations when handing loads

⚠ DANGER

Wear protective footwear.

⚠ CAUTION

Transporting passengers is strictly prohibited.

⚠ DANGER

Before picking up the load, make sure that its dimensions and weight fall within the truck specifications, as indicated in the "TECHNICAL DATA" chapter.

⚠ DANGER

The loads must be arranged so that they cannot slip or overturn and fall to the ground. In order to guarantee load stability, make sure that the load is balanced and centred on the forks.

⚠ CAUTION

Make sure that nobody stands in the truck's area of operation.

⚠ DANGER

Risk of injury and crushing for the operator and for any pedestrians! Risk of damage to the truck and the goods

During movement of the truck and the loads, especially in reverse gear, be careful not to hit any obstacles. You must maintain an adequate safety distance from obstacles (e.g. other pallets, protruding objects, shelving, pedestrians, other trucks in transit etc.).

⚠ DANGER

Risk of injury and crushing for the operator!

The operator must always remain inside the envelope of the truck.

NOTE

Further information on the general rules of truck use and picking up and depositing loads is provided in the "Safety Regulations for Industrial Forklift Use" manual attached to this manual.

Before picking up a load

Load capacity

Do not exceed the load capacity of the truck. See ⇒ Chapter "VDI data sheet for standard EXU-S 22", P. 6-101.

⚠ WARNING

Never exceed the stated maximum load capacities! Otherwise truck stability will not be ensured.

The load capacity plate must be observed. Do not mount additional weights to increase the load capacity. Do not exceed the stated maximum load capacities! Otherwise truck stability will not be ensured.

⚠ WARNING

When small parts are transported, a protective guard must be attached to prevent goods from falling onto the driver.

Taking up load

- Approach the load with caution and with as much precision as possible.
- Lower the fork arms so that they can easily be inserted into the pallet.
- Slowly insert the forks at the centre of the load to be lifted.

⚠ CAUTION

Insert the fork without bumping into either the shelving or the load.

- Insert the forks as far as possible below the load. If possible, the forks should be inserted far enough in that the load is resting against the fork carriage. The load centre of gravity must be centred between the forks.

⚠ DANGER

Pay attention to the part of the forks protruding from the load to be lifted.

Do not strike the wall, the shelving or other loads and/or objects behind the load to be picked up.

- Lift the load a few centimetres from the ground and read the "Transporting loads" section.

Transporting a load

▷

As a general rule, loads must be transported one by one (e.g. pallets). Transporting several loads at once is only authorised:

- If the safety requirements are met.
- On the orders of the supervisor in charge.

The operator must ensure that the load is properly packaged. The operator can only move loads that have been properly packaged and are safe and secure.

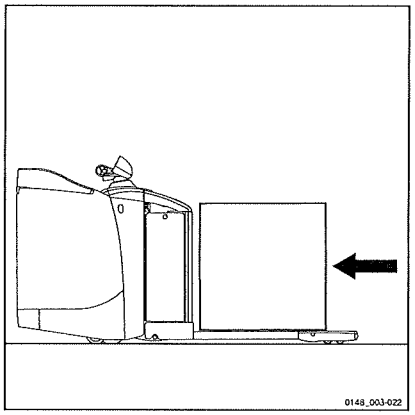
⚠ WARNING

The load must be at a height that does not obstruct the driver's view of the front, otherwise drive with the load trailing. If this is not possible, work with a guide walking beside the truck. In this case, drive the truck only at walking speed and with the utmost caution. Stop the truck immediately if you lose contact with the guide.

⚠ DANGER

Lower or raise the load until there is sufficient ground clearance.

Do not allow the load, the pallets or the container to trail along the floor.



⚠ DANGER

When travelling and transporting the load, be aware of the side clearance of the load, particularly when cornering.

Avoid hitting shelving and objects in your path.

⚠ DANGER

Danger of load tipping over

Avoid sudden starts and stops.

Approach corners slowly and carefully.

Depositing a load on the ground

- Approach the load deposit area.
- Lower the fork arms until the load is deposited in the required area, then free the forks from any contact with the pallet or container.
- Look behind you before backing the truck away
- Check that the truck's path is free of any objects, people and obstacles of any type
- Look behind you and back away very slowly to fully extract the forks from the load.

⚠ DANGER

Risk of injury and crushing for the operator! Risk of damage to the truck and the goods

During the entire load placement operation, be careful not to hit any obstacles. You must maintain an adequate safety distance from obstacles (e.g. other pallets, protruding objects, shelving etc.).

⚠ DANGER

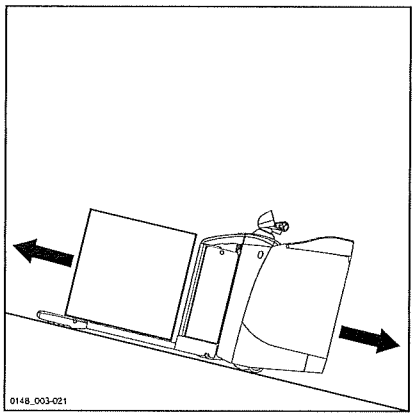
Never leave the forklift with the forks raised whether loaded or not.

NOTE

Further information on the general rules of forklift use is provided in the Rules for the Use of Industrial Vehicles Manual enclosed with this manual.

Driving up and down on an incline

When driving up and down on an incline, the load must always be carried uphill. ▷

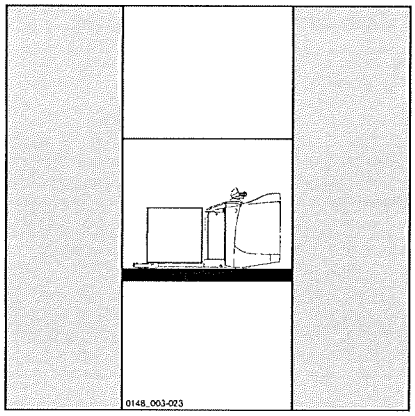


It is only permitted to drive on inclines which are identified as suitable for traffic and which can be safely negotiated according to the technical data of the truck. The driver must satisfy himself that the ground is clean and provides enough grip. It is not allowed to turn on, park on or travel across inclines. Speed must be reduced when driving down on an incline.

Entering lifts

With this truck, the driver may only enter lifts having a sufficient load capacity and for which the operator of the lift has given permission to enter.

In the lift, the truck must be secured so that no parts touch the shaft walls. ▷



The clearance to the shaft wall must at least 100 mm.

The maximum weight of the truck (dead weight without driver, the max. load.

EXU-22	3320 kg
EXU-24	3520 kg

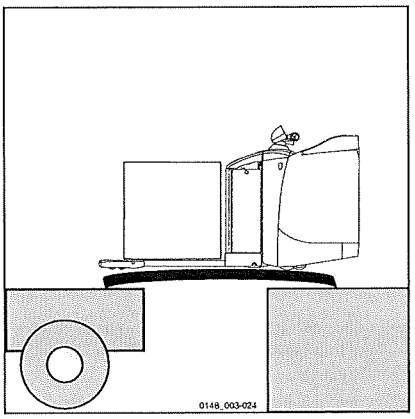
Persons travelling with the lift at the same time may enter the lift only after the truck is standing securely and they must leave the lift first.

Driving on loading bridges

Before crossing a loading bridge, the operator must make sure it is properly attached and secured and its load capacity is sufficient. ▷

Cross the loading bridge slowly and carefully. The driver must be sure that the vehicle to be entered is secured sufficiently against movement and that it can support the load of the forklift truck.

The lorry driver and lift truck operator must coordinate the departure time of the lorry.



Using the truck on inclines, loading bridges and lifts.

Driving on inclines

When driving the truck up or down inclines, you must not exceed the values for inclines indicated in the chapter "Technical data".

The operator must check that the ground is clear with a good grip.

⚠ WARNING

When driving up or down inclines, the speed of travel must be reduced.

⚠ DANGER

Risk of tipping!

When driving up or down inclines, do not turn, reverse and/or travel diagonally.

⚠ WARNING

When driving on an incline with a load, you must keep the load facing upwards.

⚠ DANGER

Risk of accident

Keep the truck at a safe distance from the edges of ramps, tailboards etc.

⚠ CAUTION

In certain cases, it is permitted to drive with the forks pointing towards the top of the incline even if the truck is not loaded.

In these cases, drive with the utmost care and avoid turning until all the wheels are on a flat surface.

⚠ DANGER

Risk of accident

Do not park on an incline: if, in the event of an emergency, you have to do so, apply the parking brake and block the wheels with chocks.

Using the truck on a lift

Using the truck on lifts is only allowed if the lift has sufficient load capacity (check the maximum weight of the truck including the traction battery) and only with appropriate authorisation.

Slowly drive the truck onto the lift load-first.

Secure the truck in the lift so that no part comes into contact with the walls of the lift. A minimum distance of 100 mm from the walls of the lift must always be observed.

⚠ WARNING

The truck must be correctly immobilised so that it cannot move inadvertently.

⚠ CAUTION

Personnel accompanying the truck onto the lift may only enter the lift once the truck is secure and must exit the lift first after transit.

Using the truck on loading bridges

⚠ DANGER

Risk of accident

Before driving on to a loading bridge, the operator must check that it has been properly fitted and secured and has sufficient load capacity.

You must drive onto the loading bridge slowly and carefully.

The operator must check that the vehicle to be loaded or unloaded is sufficiently secure so that it will not move and that it is suitable to support the strain of the truck.

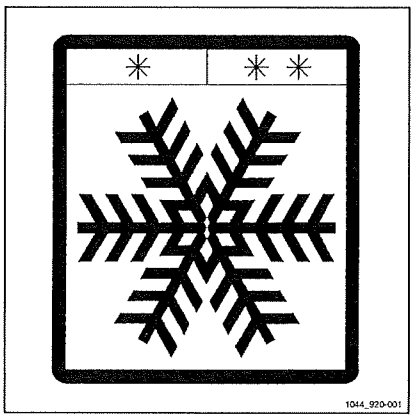
The lorry driver and the forklift truck operator must agree on the time of departure of the lorry.

Towing trailers

The forklift is not qualified to tow trailers.

Cold store usage (optional)

Designation



Your truck is fitted with special equipment for use in cold stores. It can be used for two operating ranges and carries a cold store label.

The cold store equipment for the truck consists of using specialised oils (for the hydraulic installation and the gears) suitable for cold stores.

Proper usage

Operating range 1: permanent use in areas with temperatures of –5 °C and for short periods of time down to –10 °C. Parking outside the cold store.

Operating range 2: alternating use indoors and outdoors in compliance with the rules below, temperature range from –32 °C to +40 °C. Parking outside the cold store. This use requires hydraulic oil for cold stores as given on the list of maintenance characteristics.

Use

General

The change in temperature between the cold indoors and the heat outside causes condensation. This water can freeze when the truck goes back into the cold store and jam the moving parts of the truck. This is why the length of time the truck remains in the different temperature areas given below for the two operating ranges must be strictly adhered to.

The temperature of the traction batteries must never fall to the temperature of the cold store, otherwise they will stop working.

Prior to start-up

⚠ CAUTION

The truck must be dry and at operating temperature before being used in the cold store.

- Drive the truck for approximately 5 minutes and operate the brakes several times to ensure the truck operates safely.
- Operate all the lifting functions several times. This warming up phase is required to allow the oil to reach operating temperature.

Use

Operating range 1

Permanent use in areas with temperatures of–5 °C and for short periods of time down to –10 °C.

Operating range 2

Alternating use indoors with temperatures down to –32 °C and outside with temperatures up to +25 °C for short periods of time even up to +40 °C. The truck must not leave the cold area for more than 10 minutes, because this length of time is not long enough for the formation of condensation. If the truck stays outside for longer than 10 minutes, it must remain outside for long enough to allow the condensation to run away. This generally takes at least 30 minutes.

⚠ DANGER

If the condensation freezes in the cold store, the moving parts that have become jammed must not be freed by hand.

Parking

- Always park the truck outside the cold store.

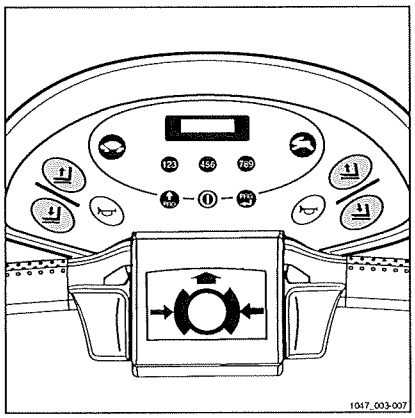
⚠ CAUTION

The batteries must not remain discharged or unused in the cold store overnight.

- Charge the battery outside the cold store and use a spare battery.

Digicode

Digicode control



Access to the electronic control is granted with three different codes:

- Driver's code
- master code
- service code

Driver's code

The driver's code consists of four digits and must be created by the Pool Manager.

It is used for the regular utilisation and operation of the truck by the driver and operator.

The knowledge of the code should be regarded as a key and must not be misused. The driver's code must not be given to other persons without express order.

The control can store a maximum of 200 driver's codes. A driver's code can be added ⇒ Chapter "Adding a driver's code", P. 4-75 or deleted ⇒ Chapter "Deleting a driver's code", P. 4-77 with the master code.

Master code

The master code consists of four digits and is reserved for the Pool Manager.

The factory setting of this code is "1234". The pool manager is advised to replace this code with a new master code ⇒ Chapter "Deleting a master code", P. 4-73 or ⇒ Chapter "Adding the master code", P. 4-72.

Like the driver's code, it is used for the regular operation and use of the truck, but it also allows the addition or deletion of a driver's code. This control can store a maximum of five master codes.

NOTE

If master codes or driver's codes are forgotten or get lost, please contact the Service Engineer, for he alone can make the stored codes visible.

Service code

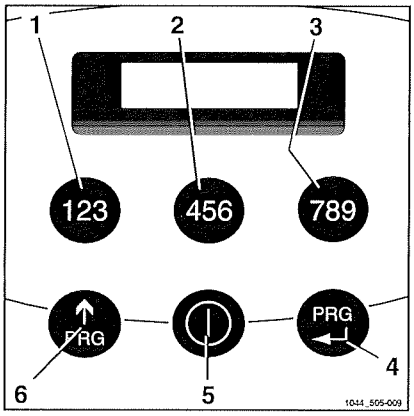
The service code is only intended for the after-sales service. It allows the diagnosis of the truck for inspections and the access to the driver's and master code.

Adding the master code

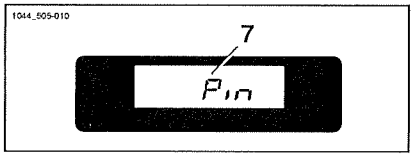
NOTE

The truck is turned off, the battery is connected.

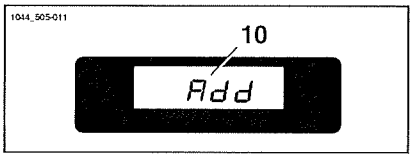
- Press the [I] (ON) button (5). ▷
- Enter a valid 4-digit master code with the buttons [123] (1), [456] (2) and [789] (3).



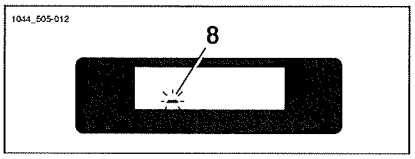
- Scroll with the [PRG ↑] (scrolling) button (6) until `Pin` (7) appears in the display. ▷



- Confirm with the [PRG ↵] (enter) button (4) until `Add` (10) appears in the display. ▷



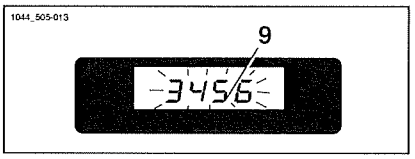
- Confirm with the PRG ↵ (enter) button (4). A flashing line cursor (8) will appear in the display.
- Enter the desired new 4-digit master code with the buttons (1), (2) or (3) as described above.



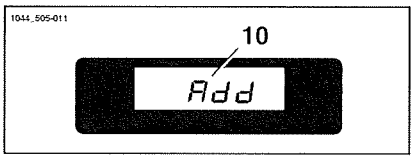
NOTE

The controller can store a maximum of five master codes. If one attempts to add a sixth master code, `Err` *will appear in the display.*

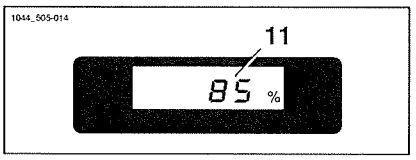
- Confirm with the PRG ↵ (enter) button (4) until the new master code (9) blinks (an example is shown).



- Then release the button. `Add` (10) will appear again in the display.



- Press the PRG ↑ (scrolling) button (6) until a normal readout (11) (eg battery charge) appears in the display. The new master code is entered.



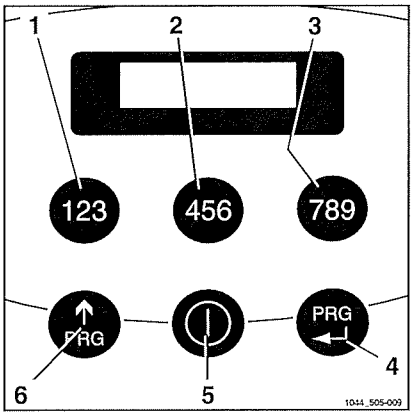
Deleting a master code

NOTE

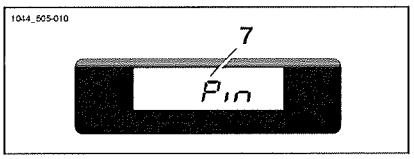
The truck is turned off, the battery is connected.

Digicode

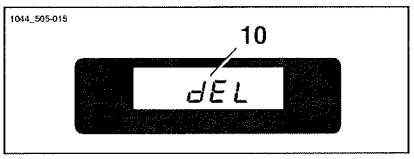
- Press the [I] (ON) button (5). ▷
- Enter the valid 4-digit master code with the buttons [123] (1), [456] (2) and [789] (3).



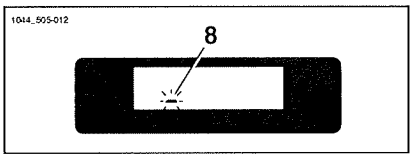
- Scroll with the [PRG ↑] (scrolling) button (6) until `Pin` (7) appears in the display. ▷



- Confirm with the [PRG ↵] (enter) button (4) until `dEL` (10) appears in the display. ▷



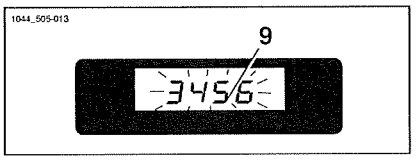
- Confirm with the [PRG ↵] (enter) button (4). A flashing line cursor (8) will appear in the display. ▷
- Enter the 4-digit master code to be deleted with the buttons (1), (2) or (3) as described above.



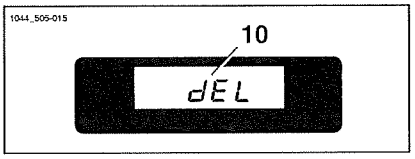
NOTE

If one attempts to delete an erroneous master code, `Err` *will appear in the display. If only one master code remains,* `None` *will appear in the display.*

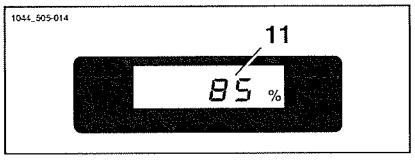
- Confirm with the PRG ↵ (enter) button (4) until the master code (9) to be deleted blinks (an example is shown). ▷



- Then release the button. dEL (10) will appear again in the display. ▷



- Press the PRG ↑ (scrolling) button (6) until a normal readout (11) (eg battery charge) appears in the display. The master code is deleted. ▷

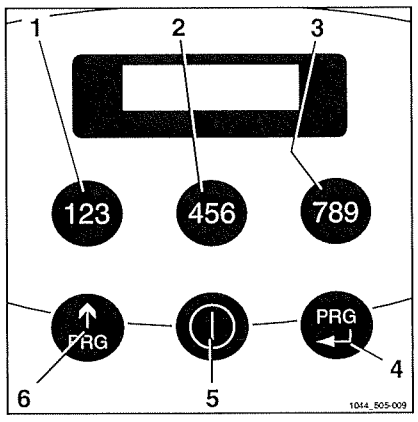


Adding a driver's code

NOTE

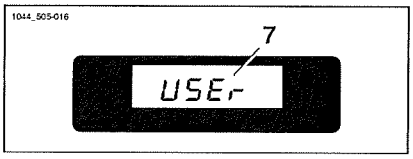
The truck is turned off, the battery is connected.

- Press the I (ON) button (5). ▷
- Enter the valid 4-digit master code with the buttons 123 (1), 456 (2) and 789 (3).

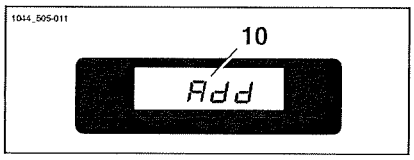


Digicode

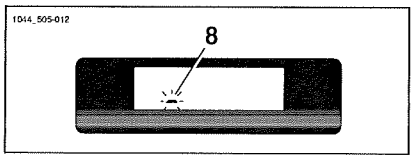
- Scroll with the PRG ↑ (scrolling) button (6) until USEr (7) appears in the display. ▷



- Confirm with the PRG ↵ (enter) button (4) until Add (10) appears in the display. ▷



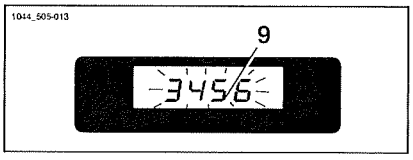
- Confirm with the PRG ↵ (enter) button (4). A flashing line cursor (8) will appear in the display. ▷
- Enter the new 4-digit driver's code with the buttons (1), (2) or (3) as described above.



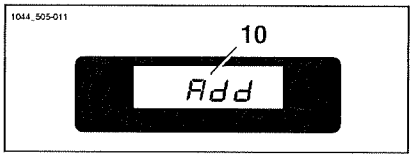
NOTE

The controller can store a maximum of 200 driver's codes. If one attempts to add a 201st driver's code, Err *will appear in the display.*

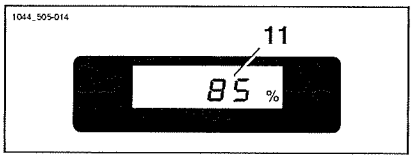
- Confirm with the PRG ↵ (enter) button (4) until the new driver's code (9) blinks (an example is shown). ▷



- Then release the button. Add (10) will appear again in the display. ▷



- Press the PRG ↑ (scrolling) button (6) until a normal readout (11) (eg battery charge) appears in the display. The new driver's code is entered. ▷



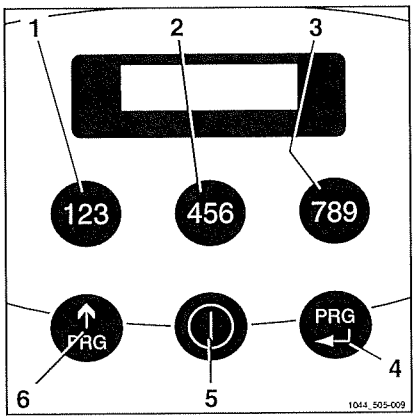
Deleting a driver's code

NOTE

The truck is turned off, the battery is connected.

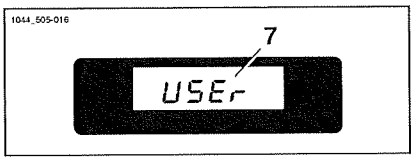
- Press the [I] (ON) button (5).
- Enter the valid 4-digit master code with the buttons [123] (1), [456] (2) and [789] (3).

▷



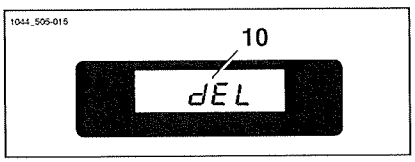
- Scroll with the [PRG ↑] (scrolling) button (6) until USEr (7) appears in the display.

▷



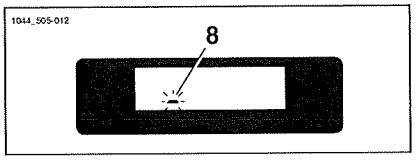
- Confirm with the [PRG ↵] (enter) button (4) until dEL (10) appears in the display.

▷



- Confirm with the [PRG ↵] (enter) button (4). A flashing line cursor (8) will appear in the display.
- Enter the 4-digit driver's code to be deleted with the buttons (2), (3) or (3) as described above.

▷

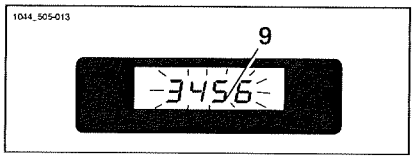


NOTE

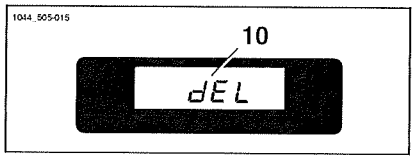
If one attempts to delete an erroneous driver's code, Err *will appear in the display.*

Digicode

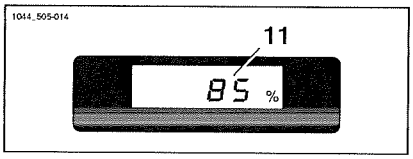
- Confirm with the PRG ↵ (enter) button (4) until the driver's code (9) to be deleted blinks (an example is shown). ▷



- Then release the button. dEL (10) will appear again in the display. ▷



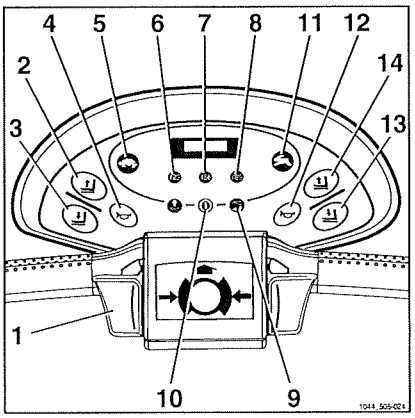
- Press the PRG ↑ (scrolling) button (6) until a normal readout (11) (eg battery charge) appears in the display. The driver's code is deleted. ▷

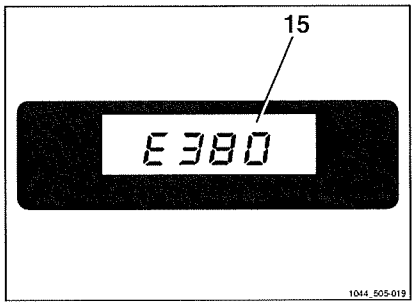


Error codes

Error code

If an error code such as E380 (15) appears, shut the truck off by pressing button (10) and repeat the start-up ⇒ Chapter "Start-up", P. 46. If the error code persists, call your Service.

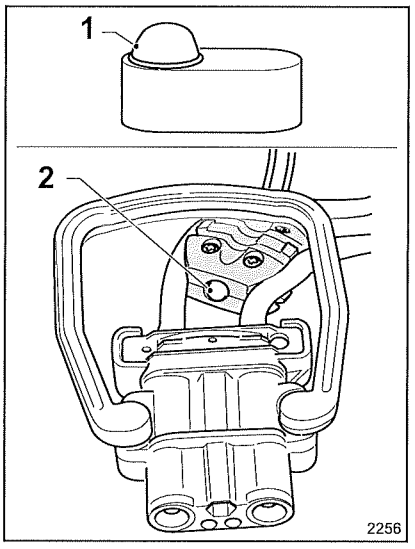




Error code	Description	Action
E350	Travel control operated at power-up.	Release the travel control (1).
E370	Left [Lift] button operated at power-up.	Release the left [Lift] button (2).
E371	Left [Lowering] button operated at power-up.	Release the left [Lowering] button (3).
E372	Left [Horn] button operated at power-up.	Release the left [Horn] button (4).
E373	[Tortoise] button operated at power-up.	Release the [Tortoise] button (5).
E374	Button [1-2-3] operated at power-up.	Release the button [1-2-3] (6).
E375	Button [4-5-6] operated at power-up.	Release the button [4-5-6] (7).
E376	Button [7-8-9] operated at power-up.	Release the button [7-8-9] (8).
E377	[PROG] button operated at power-up.	Release the button [PROG] (9).
E378	[ON] button operated at power-up.	Release the [ON] button (10).
E379	[Hare] button operated at power-up.	Release the [Hare] button (11).

Error code	Description	Action
E380	Right [Horn] button operated at power-up.	Release the right [Horn] button (12).
E381	Right [Lowering] button operated at power-up.	Release the right [Lowering] button (13).
E382	Right [Lift] button operated at power-up.	Release the right [Lift] button (14).

Battery electrolyte level indicator LED (optional)



There are two versions of the LED:

- 1) Located on the battery
- 2) Located next to the battery plug.

The LED indicates whether it is necessary to top up the distilled water in the battery.

Operation:

- If the LED (1) or (2) is green, there is a sufficient level of electrolyte in the battery. The battery must not be topped up with distilled water.
- If the LED (1) or (2) is red, there is an insufficient level of electrolyte in the battery. The battery must be topped up with distilled water.

Handling the truck in specific situations

Towing and moving

The truck cannot be towed if there is no electical power as the electric brake is applied.

If the brake of the towed truck is no longer functioning, the truck may only be towed with a fixed connection (tow bar). For towing you need a pulling vehicle with sufficient tractive and braking force for the truck being towed.

Towing without a battery

NOTE

In case of an electric failure or if no battery is installed, the brake can be released manually.

CAUTION

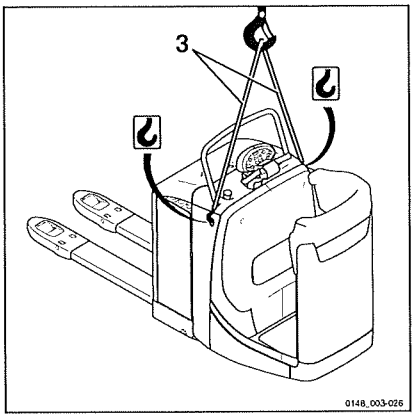
This service must be carried out by your after-sales service.

Hoisting the truck

CAUTION

Only use a lifting gear and crane with a sufficient load capacity. See the vehicle identification plate for the loading weight.

– For hoisting with a crane, always attach the slings to the points identified with the hook symbol. ▷



CAUTION

It is strictly forbidden to attach the lifting slings to the cockpit or to other points not intended for this purpose.

CAUTION

– Insert pieces of wood to prevent damage.

DANGER

Danger to life!

Do not step or stand under a suspended load.

Towing procedure

Towing the truck is only possible if there is still a power supply, otherwise the electric brake is still applied.

NOTE

The brake can be released. Please contact your after-sales service.

Handling the battery

Battery type

The pallet stacker can be equipped with different types of batteries. Observe the information on the nattery type plate. Also refer to ⇒ Chapter "Technical data", P. 99 for this purpose.

⚠ WARNING

The weight and dimensions of the battery affect truck stability.

Do not change the weight conditions when changing the battery. Do not remove nor change the position of additional weights.

Damage to cables

⚠ CAUTION

Take care not to damage the battery leads when removing and installing the battery.

Preparation

Maintenance personnel

The battery may only be changed by specially trained personnel, in accordance with the manufacturer's instructions for the battery, the battery charger and the truck. The maintenance instructions for the battery must be observed.

Fire prevention measures



⚠ WARNING

Do not smoke or use a naked flame when handling batteries. In the area designated for parking the truck to recharge the battery or battery charger, there should be no flammable materials or substances that can cause sparks within a radius of at least 2 metres. The charging area must be well ventilated. Keep a fire extinguisher at hand.

Safe parking

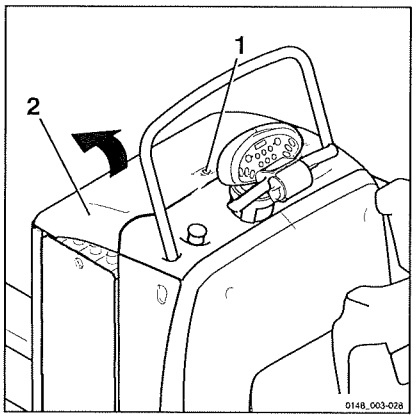
Park the truck securely before carrying out work on the battery. The truck can only be operated when the battery cover is closed and the battery outlet is inserted. If the truck is enabled for side removal of the battery, the truck can only be operated once the battery is fixed in place properly using the battery locking system.

Opening/closing the battery compartment

Opening the battery compartment

- Turn the lock (1) 90° and lift the battery cover (2).

▷



Closing the battery compartment

⚠ WARNING

Danger of crushing.

When closing the battery door, there should be nothing between battery door and edge of the frame.

- Lower the battery compartment and turn the lock 90°.

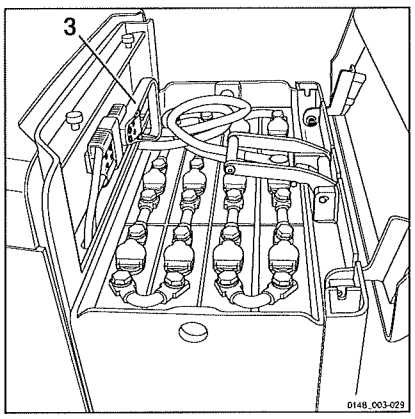
Battery charging with external battery charger

⚠ CAUTION

Deep discharges damage the battery.

- The battery must be charged at once.
- Park the truck safely, refer to ⇒ Chapter "Safe parking of the truck", P. 4-87.
- Open the battery compartment.
- Before charging, inspect the battery leads and charging cables for damage and replace them, if required.

– Disconnect the battery connector (3). ▷



⚠ WARNING

Switch the truck and charger off before disconnecting the male and female connectors.

– Connect the battery connector to the charger connector.

NOTE

Observe the information in the operating instructions for your battery and battery charger (equalising charge).

⚠ WARNING

Risk of damage, shorting and explosion.

Do not place any metallic objects or tools on the battery. No naked lights, no smoking permitted.

⚠ WARNING

The electrolyte (diluted sulphuric acid) is poisonous and caustic!

Observe the safety precautions for handling battery acid.

⚠ WARNING

Explosive gases are released during charging.

– Ensure that the room ventilation is adequate.

Changing the battery

There are two possibilities of replacing the battery:

– Vertically with a lifting tackle (all trucks).

– from the side using a battery trolley (optional - only for EXU 18 and 20 with side door).

In order to avoid short-circuits, it is recommended that batteries with polar terminals or unprotected connections be covered with a rubber mat.

⚠ DANGER

Danger to life!

No-one should be under a raised load.

Changing the battery with crane

Maintain a safe distance for battery removal to avoid damaging the truck.

- Open the battery compartment (1). ▷
- Unplug the battery connector.

In order to avoid short-circuits,it is recommended that batteries with polar terminals or unprotected connections be covered with a rubber mat.

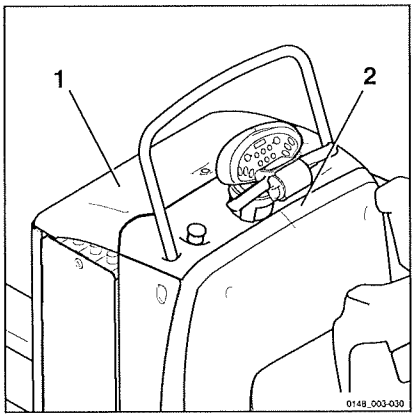


⚠ DANGER

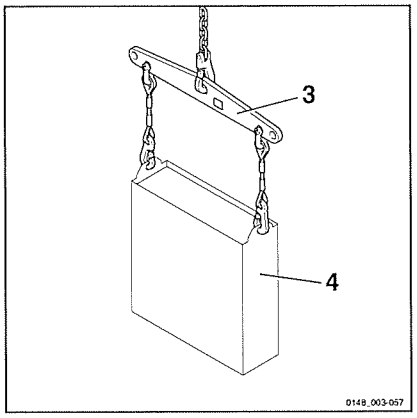
Danger to life!

No-one should be under a raised load.

- Unlock the battery.



- Attach the lifting device (3) correctly to the battery (4) (see the lifting device user manual), then remove it from the truck. ▷
- The lifting device must exercise vertical traction in order to avoid the boot being damaged. The hooks must be placed so that they cannot fall on the battery cells when the lifting device is slackened.
- Installation and connection of the charged battery is the reverse of removal. Make sure lateral cover (2) is properly put in place.



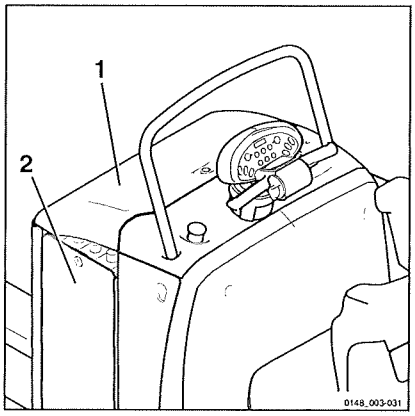
Changing the battery with lateral removal trolley

⚠ CAUTION

When removing or installing the battery keep your hands away from chassis and battery! There is an increased risk of squeezing your hands!

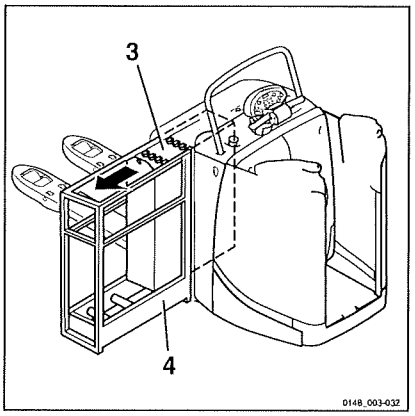
- Open the battery cover (1).
- Remove the side cover (2).

▷



- Place a suitable trolley (4) next to the battery.
- Position yourself to the side opposite to the side cover and push the battery (3) onto the removal trolley (4).
- To install the new battery position yourself opposite to the side cover and pull the battery towards you into the battery compartment.
- Put the side cover (2) in place.

▷



Decommissioning

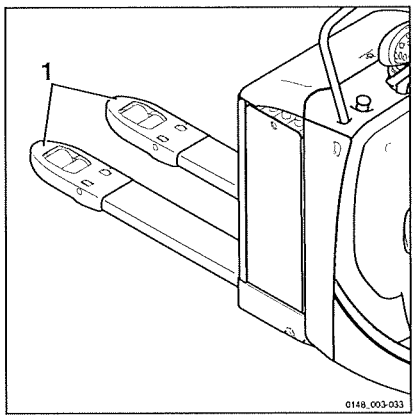
Safe parking of the truck

– Park the truck in a dry, clean and well-ventilated area.

⚠ WARNING

Do not park the truck on gradients. In exceptional cases, secure the truck with chocks.

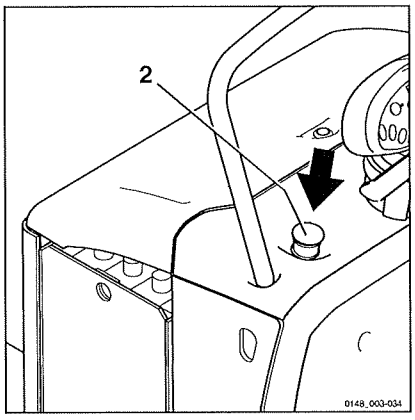
– Lower the forks (1) on the ground. ▷



⚠ WARNING

– Before leaving the truck, lower the load fully.

– Depress the emergency isolator button (2). ▷



ⓘ NOTE

If you leave the truck without disconnecting the battery connector, the control is shut off automatically after approx. 15 minutes.

The truck can only be operated again after re-entering the driver code.

Storage

Measures when storing the stacker for a longer period of time

If the stacker is to be held in storage for a longer longer period of time, carry out the following corrosion protection measures. If the stacker is to be stored for over two months, park it in a clean and dry storage area, which is also well ventilated and frost-free. Also perform the following measures:

Measures before taking out of operation

- Clean the stacker thoroughly.
- Raise the forks several times as far as possible.
- Check the hydraulic oil level, adding oil if necessary.
- Lower the forks on to an appropriate support, for example a pallet, until the lift chains are relaxed.
- Coat all blank, moving parts with a thin film of oil or grease.
- Oil all moving parts and joints.
- Check the condition and electrolyte density of the battery, service the battery according to the instructions of the manufacturer. (Follow the battery manufacturer's instructions.)
- Spray all naked electric contacts with a suitable contact spray.

⚠ CAUTION

- Block up the truck so that the wheels are clear of the ground. This will prevent a permanent deformation of the tyres.
- Cover the pallet stacker with a cotton blanket and protect it against dust.

⚠ CAUTION

We do not recommend using a plastic foil as this enhances the formation of condensate.

If the truck is to be taken out of service for an even longer period, contact your after-sales service.

Taking back into service after storage

If the pallet stacker has been out of service for over six months, it must be inspected carefully before it is taken back into service. The inspection, similarly to the safety inspection, should also include safety-related points on the truck.

- Clean the stacker thoroughly.
- Oil all moving parts and joints.
- Check the condition and acid density of the battery, charging it if required.
- Check the hydraulic oil for condensate and renew the oil, if necessary.
- Perform the services as before initial operation.
- Take the pallet stacker into operation.
- During the next commissioning, check in particular:

• Traction drive, control, steering
• Brake (service brake, parking brake)
• Lifting device

5

Maintenance

General Information

To keep your forklift in good condition, carry out the servicing indicated regularly, within the times indicated and using the consumption materials provided for that purpose, as specified on the following pages. Please make sure that you keep a record of work done; this is the only way for the guarantee to remain valid.

Maintenance is divided into:

- Regular Service (scheduled by the user)
- Planned maintenance (to be performed by the service network authorised by the manufacturer)

⚠ DANGER

Planned maintenance and repairs must be performed by the service network authorised by the manufacturer in order to keep the machine in perfect condition and compliant with technical specifications.

ⓘ NOTE

Contact the authorised service network for a maintenance contract appropriate to your forklift.

⚠ CAUTION

Maintenance intervals are defined for standard use. In the following cases, it is necessary to reduce the interval between the various scheduled maintenance operations: in the event of use in dusty or salty environments, extremely high or low ambient temperatures, high levels of air humidity, particularly intense and heavy-duty uses, and specific national regulations for trucks or individual components.

Operations Preliminary to Maintenance

Do the following before performing maintenance operations:

- Place the truck on a flat surface and make sure that it cannot move accidentally.
- Lower the forks fully.
- Switch off the vehicle.
- Press the emergency stop button.

⚠ DANGER

Before performing any intervention on the electric system, disconnect the battery outlet from the relative plug.

Scheduled maintenance

Summary table of maintenance operations EXU-S

Operations	Intervals in hours			Completed
	1000 (a)	2000 (b)	5000 (c)	v
Checks and tests prior to commissioning	•			
Check the oil level in the hydraulic system and the transmission gear	•			
Test the insulation between the chassis and the electric motors	•			
Test the insulation between the chassis and the electronic control	•			
Check the condition of the battery and check that it is correctly mounted	•			
Check the condition of the electrical system (wiring harnesses and fuses) and that it is correctly mounted	•			
Check for wear on the lift motor brushes and replace if necessary	•			
Check the truck braking	•			
Check and adjust the electromagnetic brake	•			
Check that the forks are in good condition	•			
Check the condition of the piping	•			
Check for oil leakages from cylinders and hydraulic connectors	•			
Tyre wear check	•			
Check the correct mounting of the electrical steering pinion gear	•			
Check wheel tightness	•			
Change the hydraulic oil and filter		•		
Service the transmission gear			•	

1000 (a) = To be repeated every 1000 hours (for example at 1000, 2000, 3000, 4000, 5000 etc.) or at least every 12 months (whichever comes first).

2000 (b) = To be repeated every 2000 hours. For example at 2000, 4000, 6000, 8000, 10,000 etc.)

5000 [c] = To be repeated every 5000 hours. For example at 5000, 10,000, 15,000, 20,000 etc.)

ENVIRONMENT NOTE

During maintenance operations, follow the instructions provided in the section "Safety regulations relative to operating materials" in "Chapter 2".

Maintenance as required

Cleaning the Forklift

Cleaning depends on the type of use and the workplace. Should the truck come into contact with highly aggressive elements such as salt water, fertilizers, chemical products, cement, etc., it should be cleaned as carefully as possible after every work cycle. It is preferable to use cold compressed air and detergents.

Use water-dampened rags to clean the parts of the body.

⚠ CAUTION

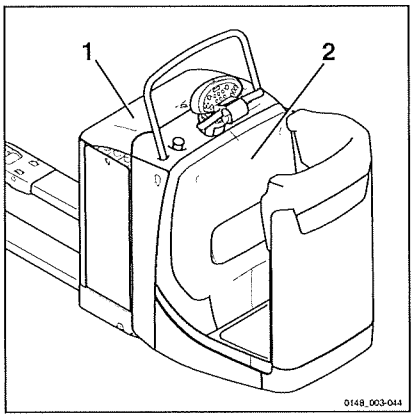
Do not clean the truck with direct jets of water; DO NOT use solvents and petrols that could damage parts of the truck.

Fuses

⚠ CAUTION

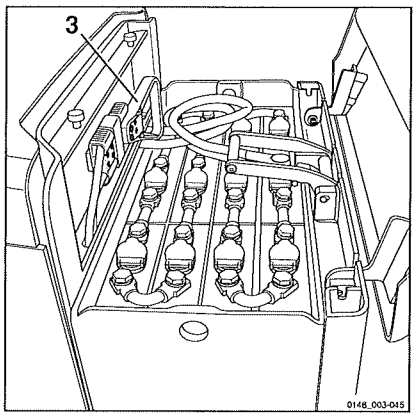
Before performing any work on the electrical installation, remove any power from the system by disconnecting the battery connector.

– Open the battery cover (1). ▷



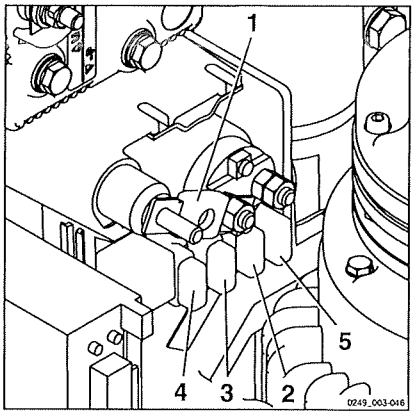
Maintenance as required

- Disconnect the battery connector (3). ▷
- Remove the bonnet(2) to gain access to the fuse carrier.



The following fuses are located on the fuse carrier: ▷

1 Main fuse	400 A
2 Fuse F3	7.5 A
3 Fuse F4	7.5 A
4 Fuse F5	20 A
5 Fuse F8 (Option)	3 A



Decommissioning

General Information

The operations to be performed for "**Temporary decommissioning**" and "**Permanent decommissioning**" are listed in this chapter.

Forklift Towing

The forklift may not be towed in the case of breakdown.

The forklift must be lifted with due caution, as described on the preceding pages.

Temporary Putting Out of Commission

The following operations must be performed when the forklift is not going to be used for a long time:

- Clean the forklift as indicated in the "Maintenance" chapter and put it in a dust-free and dry room. -
- Lower the forks.
- Lightly grease all of the unpainted parts with oil or grease.
- Perform the lubrication operations indicated in the maintenance chapter.
- Remove the battery and put it in a room where there is no danger of freezing. Charge the battery at least once a month.
- Raise the forklift so that the wheels do not touch the ground; otherwise, the wheels will become flat at the point of contact with the floor.
- Cover the forklift with a NON-plastic sheet.

Checks and Inspections After a Long Period of Inactivity

⚠ DANGER

Perform the following operations before using the forklift:

- Clean forklift truck thoroughly.
- Check the battery charge level and re-assemble it in the forklift, making sure to spread Vaseline on the terminals.
- Lubricate all of the parts provided with lubricating nipples and the chains.
- Carry out the fluid level checks.
- Perform all of the functional maneuvers of the forklift and of its safety devices both loaded and unloaded.

⚠ DANGER

Follow the instructions provided in the maintenance chapter for the operations indicated previously.

Permanent Putting Out of Commission (Demolition)

The forklift must be demolished in compliance with local legislation. Contact the authorised service network or authorised companies to scrap the forklift according to local legislation.

ENVIRONMENT NOTE

In particular, batteries, fluids (oils, fuels, lubricants, etc, electrical and electronic components and rubber components must be disposed of in compliance with specific local legislation for each type of material.

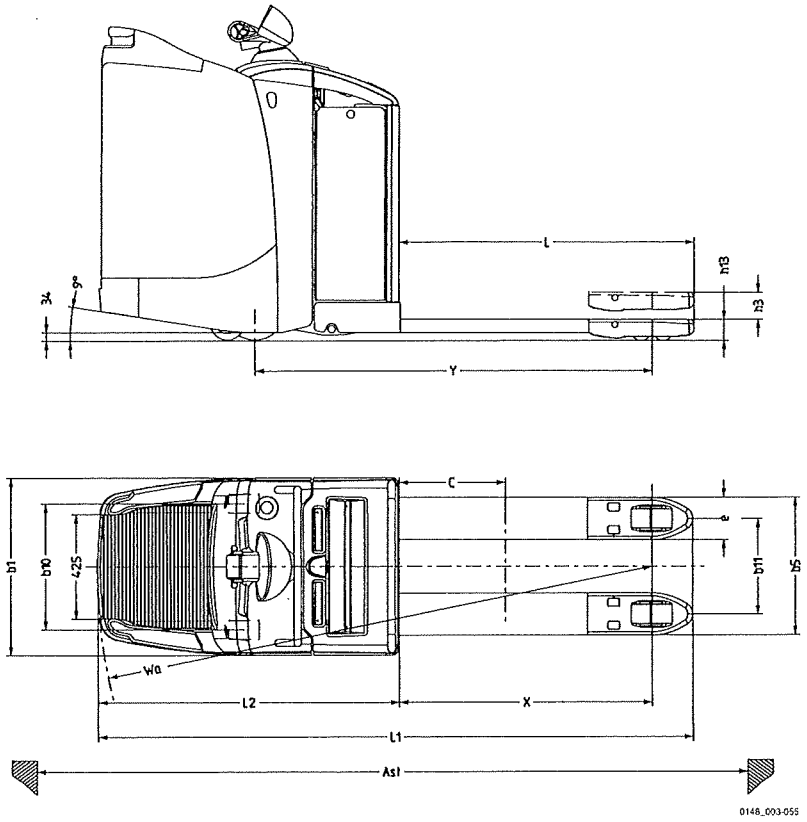
⚠ DANGER

Disassembly of the forklift for scrapping is extremely hazardous.

6

Technical data

Dimensions



VDI data sheet for standard EXU-S 22

Characteristics

			EXU-S 22	EXU-S 22
Type of platform			Rear access	Side access
Drive: electric, diesel, petrol, LPG.			Electric	
Driving mode: manual, pedestrian, ride-on standing, ride-on seated, order picking			Standing	
Rated/load capacity		Q (kg)	2200	
Nominal load centre of gravity		c (mm)	600 (1)	
Load distance		x (mm)	1023 (2)	
Centre-to-centre distance		y (mm)	1617 (2)	

Weights

			EXU-S 22	EXU-S 22
Tare weight	including battery	kg	1120	
Pressure on the axles, with load	drive side/load side	kg	1820/1500	
Loads on the axles, without load	drive side/load side	kg	210/910	

Wheels and tyres

			EXU-S 22	EXU-S 22
Tyres			Polyurethane	
Wheel size	drive side	mm	Ø250 x 100	
Wheel size	load side	mm	Ø85 x 80	
Pivoting wheels	drive side	mm	Ø140 x 54	

VDI data sheet for standard EXU-S 22

			EXU-S 22	EXU-S 22
Number of wheels (x = drive)	drive side/load side		4/1x + 2	
Track	drive side	b10 (mm)	348/388/498	
Track	load side	b11 (mm)	470	

Dimensions

			EXU-S 22	EXU-S 22
Lift		h3 (mm)	130	
Step plate height		h7 (mm)	225	
Height of cockpit in operating position	min/max	h14 (mm)	1030	
Height of forks when lowered		h13 (mm)	85	
Overall length unladen		l1 [mm]	2415 (3)	2487 (3)
Length of the drive unit		l2 [mm]	1225 (3)	1297 (3)
Overall width		b1 (mm)	720	
Dimensions of forks		s/e/l (mm)	55 (78 max)/172/1190	
Fork arm external spread		b5 (mm)	520/560/670	
Ground clearance, loaded, under mast		m2 (mm)	30	
Aisle width for pallet 800 x 1200 lengthwise (b12 x l6)		Ast (mm)	2821 (5)	2890 (5)
Aisle width for pallet 1000 x 1200 lengthwise (b12 x l6)		Ast (mm)	2791 (5) (6)	2861 (5) (6)
Bending radius		Wa (mm)	2125 (5)	2195 (5)

Performance data

			EXU-S 22	EXU-S 22
Driving speed	with/without load	km/h	10/12	
Lifting speed	with/without load	m/s	0.043/0.062	
Lowering speed	with/without load	m/s	0.068/0.065	
Surmountable gradient	with/without load	%	10%/15%	
Acceleration time (over 10 metres)	laden/unladen	s	6.2/4.6	
Service brake			electromagnetic	

Electric motors

			EXU-S 22	EXU-S 22
Travel motor, rating S2 = 60min		kW	3.0	
Lifting motor, rating S3 = 15%		kW	2.2	
Battery in accordance with DIN 43 531/35/36 A, B, C, no			no	
Battery voltage/nominal capacity K_5		V/Ah	24/465	
Battery weight (+/- 5%)		kg	405	
Energy consumption according to VDI cycle		kWh/h	1.05	

Other

			EXU-S 22	EXU-S 22
Speed control			AC-controller	
Average noise level measured at the driver's ear		dB (A)	< 65	

(1) For forks with l = 1190 mm

(2) With forks lowered. For the value with forks raised, deduct 113 mm

(3) With compartment 71: +85 mm

(4) Dimensions relating to fork length marked with an asterisk

(5) With forks raised

(6) Values calculated with W_a and fork length l = 990 mm

Edition dated 22/10/2015

VDI data sheet for standard EXU-S 24

Characteristics

			EXU-S 24	EXU-S 24
Type of platform			Rear access	Side access
Drive: electric, diesel, petrol, LPG			Electric	
Driving mode: manual, pedestrian, ride-on standing, ride-on seated, order picking			Standing	
Rated/load capacity		Q (kg)	2400	
Nominal load centre of gravity		c (mm)	600 (1)	
Load distance		x (mm)	1005 (2)	
Centre-to-centre distance		y (mm)	1599 (2)	

Weights

			EXU-S 24	EXU-S 24
Tare weight	including battery	kg	1150	
Pressure on the axles, with load	drive side/load side	kg	2233/1317	
Loads on the axles, without load	drive side/load side	kg	232/918	

Wheels and tyres

			EXU-S 24	EXU-S 24
Tyres			Polyurethane	
Wheel size	drive side	mm	Ø250 x 100	
Wheel size	load side	mm	Ø85 x 80	
Pivoting wheels	drive side	mm	Ø140 x 54	
Number of wheels (x = drive)	drive side/load side		4/1x + 2	

			EXU-S 24	EXU-S 24
Track	drive side	b10 (mm)	348/388/498	
Track	load side	b11 (mm)	470	

Dimensions

			EXU-S 24	EXU-S 24
Lift		h3 (mm)	130	
Height of cockpit in operating position	min/max	h14 (mm)	1030	
Step plate height	min/max	h7 (mm)	225	
Height of forks when lowered		h13 (mm)	85	
Overall length unladen		l1 [mm]	2415 (3)	2487 (3)
Length of the drive unit		l2 [mm]	1225 (3)	1297 (3)
Overall width		b1 (mm)	720	
Dimensions of forks		s/e/l (mm)	61 (78 max)/172 (4)/1190	
Fork arm external spread		b5 (mm)	520/560/670	
Ground clearance, loaded, under mast		m2 (mm)	24 (4)	
Aisle width for pallet 800 x 1200 lengthwise (b12 x l6)		Ast (mm)	2814 (5)	2883 (5)
Aisle width for pallet 1000 x 1200 lengthwise (b12 x l6)		Ast (mm)	2781 (5) (6)	2851 (5) (6)
Bending radius		Wa (mm)	2107 (5)	2177 (5)

Performance data

			EXU-S 24	EXU-S 24
Driving speed	with/without load	km/h	9/12	
Lifting speed	with/without load	m/s	0.043/0.062	
Lowering speed	with/without load	m/s	0.068/0.065	
Surmountable gradient	with/without load	%	9.3%/15%	

			EXU-S 24	EXU-S 24
Acceleration time (over 10 metres)	laden/unladen	s	6.3/4.8	
Service brake			electromagnetic	

Electric motors

			EXU-S 24	EXU-S 24
Travel motor, rating S2 = 60min		kW	3.0	
Lifting motor, rating S3 = 15%		kW	2.2	
Battery in accordance with DIN 43 531/35/36 A, B, C, no			no	
Battery voltage/nominal capacity K_5		V/Ah	24/465	
Battery weight (+/- 5%)		kg (+/- 5%)	405	
Energy consumption according to VDI cycle		kWh/h	1.07	

Other

			EXU-S 24	EXU-S 24
Speed control			AC-controller	
Average noise level measured at the driver's ear		dB (A)	< 65	

(1) For forks with l = 1190 mm

(2) With forks lowered. For the value with forks raised, deduct 113 mm

(3) With compartment 71: +85 mm

(4) Dimensions relating to fork length marked with an asterisk

(5) With forks raised

(6) Values calculated with W_a and fork length l = 990 mm

Edition dated 22/10/2015

Turning radius and load distance EXU-S

EXU-S 22

Turning radius (Wa) and load distance (x) with forks lowered (a)					
		Battery compartment 70 (3PzS)		Battery compartment 71 (4PzS)	
		345 - 465 Ah		560 - 620 Ah	
Forks	Load	Rear	Side	Rear	Side
length	distance	access	access	access	access
l	x	Wa	Wa	Wa	Wa
mm	mm	mm	mm	mm	mm
990	823	2038	2108	2123	2193
1190	1023	2238	2308	2323	2393
1600	1433	2648	2718	2733	2803
2390 c	not available				
2390 d					

EXU-S 24

Turning radius (Wa) and load distance (x) with forks lowered (a)					
		Battery compartment 70 (3PzS)		Battery compartment 71 (4PzS)	
		345 - 465 Ah		560 - 620 Ah	
Fork	Load	Rear	Side	Rear	Side
length	distance	access	access	access	access
l	x	Wa	Wa	Wa	Wa
mm	mm	mm	mm	mm	mm
990	805	2020	2090	2105	2175
1190	1005	2220	2290	2305	2375
1450 b	1135	2350	2420	-	-
1600	1415	2630	2700	2715	2785
2390 c	2205	3420	3490	3505	3575
2390 d	1847	3062	3132	3147	3217

Legend:

a = for value with forks raised deduct 113 mm

b = available only when b5 = 560 mm

c = useful for transporting three Euro pallets on the short side

d = useful for transporting two Euro pallets on the long side

Wheels and tyres

Approved types of wheels

Only the wheel types listed in the parts catalogue may be used.

Supply table EXU-S EXU-SF

	Standard version	Quantity
Traction transmission gear oil	FUCHS Renolin PG 220	1.1 litres
Lifting hydraulics oil	Olio idraulico HLF 32	2.5 litres
Oil for pivot wheels (only if the "STILL DRIVEspeed" version is present)	AP31 hydraulic oil	-
Generic lubricant	Tutela MP 02	-

	Cold storage version	Quantity
Traction transmission gear oil	FUCHS Renolin PG 220	1.1 litres
Lifting hydraulics oil	EQUIVIS XV32	2.5 litres
Oil for pivot wheels (only if the "STILL DRIVEspeed" version is present)	AP31 hydraulic oil	-
Generic lubricant	Tutela EP2	-

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STILL GmbH

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